NOTRE DAME UNIVERSITY BANGLADESH

Machine Learning Lab Final Report

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Submitted by:

Name: Istiak Alam

ID: 0692230005101005

Batch: CSE-20

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Submitted to:

A. H. M. Saiful Islam

Chairman, Department of CSE

Notre Dame University Bangladesh

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Abstract

Machine Learning has become an essential component in modern data-driven systems, enabling automated decision-making and predictive analytics. This lab report presents a comprehensive overview of fundamental machine learning techniques implemented across seven laboratory sessions. The experiments cover key areas including data preprocessing, regression analysis, classification algorithms, encoding techniques, probabilistic modeling, and model evaluation.

Throughout the labs, various algorithms such as Linear Regression, K-Nearest Neighbors (KNN), Naive Bayes, Logistic Regression, Support Vector Machine (SVM), and Decision Trees were implemented and analyzed. Additionally, techniques like label encoding, one-hot encoding, and confusion matrix evaluation were explored to enhance model performance and interpretability.

The objective of this report is to demonstrate practical understanding of machine learning concepts by applying them to real-world datasets and evaluating their effectiveness through systematic experimentation.

Objective

The primary objective of this lab report is to develop a strong practical foundation in machine learning by implementing and analyzing various algorithms and techniques. The specific goals include:

- To understand and perform data preprocessing for preparing datasets.
- To implement Linear and Multiple Linear Regression for predictive modeling.
- To apply K-Nearest Neighbors (KNN) for both classification and regression tasks.
- To learn and apply encoding techniques such as label encoding and one-hot encoding.
- To implement probabilistic models using the Naive Bayes algorithm.
- To understand classification techniques using Logistic Regression and Support Vector Machines (SVM).
- To implement Decision Trees and evaluate model performance using Confusion Matrix.
- To compare different machine learning models based on accuracy and prediction performance.

Introduction

Machine Learning is a subfield of artificial intelligence that focuses on building systems capable of learning from data and improving their performance without explicit programming. It plays a vital role in various applications such as prediction, classification, recommendation systems, and pattern recognition.

In this lab course, several fundamental machine learning techniques were explored through practical implementation. The process began with data preprocessing, which involves cleaning, transforming, and preparing raw data for analysis. Proper preprocessing ensures that models perform efficiently and produce reliable results.

Following this, regression techniques such as Linear Regression and Multiple Linear Regression were implemented to understand relationships between variables and perform prediction tasks. Classification techniques including K-Nearest Neighbors, Logistic Regression, Support Vector Machine, and Decision Trees were studied to categorize data into distinct classes.

Additionally, encoding methods like label encoding and one-hot encoding were applied to handle categorical data effectively. The Naive Bayes algorithm introduced probabilistic classification concepts, while the confusion matrix provided a structured approach to evaluating model performance.

Overall, this report reflects a systematic progression from basic data handling to advanced machine learning model implementation and evaluation.

Lab 1 Data Preprocessing

Objective

The objective of this experiment is to familiarize with basic data handling and preprocessing techniques required before applying Machine Learning algorithms. This lab focuses on loading a dataset, inspecting its structure, handling categorical variables, and converting them into numerical representations suitable for model training.

Dataset Description

The dataset used in this experiment is `bank.csv`, which contains customer information related to a banking institution. The dataset includes demographic details, financial attributes, and campaign-related information. The target variable indicates whether a customer subscribed to a term deposit. The main objective is to load a CSV dataset into a Pandas DataFrame and inspect the initial records.

1.1 Importing Required Libraries

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Explanation

Machine Learning workflows rely on multiple Python libraries, each serving a specific purpose. NumPy provides efficient data structures and mathematical functions for numerical computation. Pandas enables loading, cleaning, and manipulating structured datasets. Matplotlib is used to generate basic plots and graphs, while Seaborn builds on Matplotlib to produce more informative and visually appealing statistical visualizations. Importing these libraries prepares the environment for data analysis and model development.

Output

No visible output is produced if the libraries are successfully imported. If any library is missing from the environment, an error message such as `ModuleNotFoundError` is displayed.

1.2 Loading CSV Data

[2]:
```
import pandas as pd

df = pd.read_csv("bank.csv")
df.head()
```

Explanation

Pandas provides the `read_csv()` function to load structured data into a DataFrame. Initially, the dataset was loaded without specifying the delimiter, which resulted in improper column parsing. This step helps identify formatting issues in raw data.

Output

The output displays the first five records of the dataset, revealing incorrect column separation due to the delimiter mismatch.

[2]:
```
age;"job";"marital";"education";"default";"balance";"housing";"loan";
"contact";"day";"month";"duration";"campaign";"pdays";"previous";"poutcome";
"y"
0  30;"unemployed";"married";"primary";"no";1787;…
1  33;"services";"married";"secondary";"no";4789;…
2  35;"management";"single";"tertiary";"no";1350;…
3  30;"management";"married";"tertiary";"no";1476…
4  59;"blue-collar";"married";"secondary";"no";0;…
```

1.3 Loading the Dataset and Displaying Records

[3]:
```
df = pd.read_csv("bank.csv", sep = ';')
df.head()
```

Explanation

The dataset `bank.csv` uses a semicolon (`;`) as its field separator instead of the default comma. Therefore, the delimiter is explicitly specified while loading the file. The `read_csv()` function stores the data in a Pandas DataFrame, which is a tabular data structure. The `head()` function is then used to display the first five records of the dataset, allowing quick verification of data correctness and column structure.

Output

The output displays the first five rows of the dataset along with all column names, confirming that the data has been loaded correctly into the DataFrame.

[3]:

	age	job	marital	education	default	balance	housing	loan	\
0	30	unemployed	married	primary	no	1787	no	no	
1	33	services	married	secondary	no	4789	yes	yes	
2	35	management	single	tertiary	no	1350	yes	no	
3	30	management	married	tertiary	no	1476	yes	yes	
4	59	blue-collar	married	secondary	no	0	yes	no	

	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	cellular	19	oct	79	1	-1	0	unknown	no
1	cellular	11	may	220	1	339	4	failure	no
2	cellular	16	apr	185	1	330	1	failure	no
3	unknown	3	jun	199	4	-1	0	unknown	no
4	unknown	5	may	226	1	-1	0	unknown	no

1.4 Displaying the Last Records of the Dataset

[4]:
```
df.tail()
```

Explanation

The `tail()` function in Pandas is used to display the last few entries of a DataFrame. By default, it returns the final five rows. This operation is useful for verifying that the dataset has been loaded completely and for checking any anomalies or missing values at the end of the dataset.

Output

The output displays the last five rows of the dataset along with all corresponding column values.

[4]:

	age	job	marital	education	default	balance	housing	loan	\
4516	33	services	married	secondary	no	-333	yes	no	
4517	57	self-employed	married	tertiary	yes	-3313	yes	yes	
4518	57	technician	married	secondary	no	295	no	no	
4519	28	blue-collar	married	secondary	no	1137	no	no	
4520	44	entrepreneur	single	tertiary	no	1136	yes	yes	

	contact	day	month	duration	campaign	pdays	previous	poutcome	y
4516	cellular	30	jul	329	5	-1	0	unknown	no
4517	unknown	9	may	153	1	-1	0	unknown	no
4518	cellular	19	aug	151	11	-1	0	unknown	no
4519	cellular	6	feb	129	4	211	3	other	no
4520	cellular	3	apr	345	2	249	7	other	no

1.5 Encoding Marital Status Attribute

[5]:
```
def replace_marital(val):
  if val == 'single':
    return 0
  else:
    return 1

df['marital'] = df['marital'].apply(replace_marital)
df.head()
```

Explanation

In this step, the categorical attribute `marital` is converted into a numerical format to make it suitable for Machine Learning algorithms. A user-defined function named `replace_marital` is created, which assigns the value 0 to customers with marital status `single` and the value 1 to all other categories. The `apply()` function is then used to apply this transformation to the entire `marital` column of the dataset. This process simplifies categorical data while preserving relevant information.

Output

The output displays the first five rows of the dataset where the `marital` column is successfully transformed into numerical values. Customers with marital status `single` are represented by 0, while all other marital statuses are represented by 1.

[5]:

	age	job	marital	education	default	balance	housing	loan	\
0	30	unemployed	1	primary	no	1787	no	no	
1	33	services	1	secondary	no	4789	yes	yes	
2	35	management	0	tertiary	no	1350	yes	no	
3	30	management	1	tertiary	no	1476	yes	yes	
4	59	blue-collar	1	secondary	no	0	yes	no	

	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	cellular	19	oct	79	1	-1	0	unknown	no
1	cellular	11	may	220	1	339	4	failure	no
2	cellular	16	apr	185	1	330	1	failure	no
3	unknown	3	jun	199	4	-1	0	unknown	no
4	unknown	5	may	226	1	-1	0	unknown	no

1.6 Encoding Housing Loan Attribute

[6]:
```
df["housing"] = df["housing"].map({"yes": 1, "no": 0}.get)
df.head()
```

Explanation

In this step, the categorical attribute `housing` is converted into a numerical format using the `map()` function. The values `yes` and `no` are mapped to 1 and 0 respectively. This conversion is necessary because Machine Learning algorithms require numerical input features. The `get` method ensures safe mapping of values within the column.

Output

The output displays the first five rows of the dataset where the `housing` column has been successfully transformed into numerical values. A value of 1 indicates the presence of a housing loan, while 0 indicates the absence of a housing loan.

[6]:

	age	job	marital	education	default	balance	housing	loan	\
0	30	unemployed	1	primary	no	1787	0	no	
1	33	services	1	secondary	no	4789	1	yes	
2	35	management	0	tertiary	no	1350	1	no	
3	30	management	1	tertiary	no	1476	1	yes	

```
4   59  blue-collar        1  secondary      no        0        1   no

    contact  day month  duration  campaign  pdays  previous poutcome   y
0  cellular   19   oct        79         1     -1         0  unknown  no
1  cellular   11   may       220         1    339         4  failure  no
2  cellular   16   apr       185         1    330         1  failure  no
3   unknown    3   jun       199         4     -1         0  unknown  no
4   unknown    5   may       226         1     -1         0  unknown  no
```

1.7 Encoding Loan Attribute

[7]:
```
df["loan"] = df["loan"].replace({"yes": 1, "no": 0})
df.head()
```

Explanation

In this step, the categorical attribute `loan` is converted into numerical form to ensure compatibility with Machine Learning algorithms. The `replace()` function is used to map the value `yes` to 1 and `no` to 0. This binary encoding simplifies the data representation while retaining the original meaning of the attribute.

Output

The output displays the first five rows of the dataset, where the `loan` column is successfully transformed into numerical values. A value of 1 indicates the presence of a personal loan, while 0 indicates the absence of a loan.

[7]:
```
   age          job  marital  education default  balance  housing  loan  \
0   30   unemployed        1    primary      no     1787        0     0
1   33     services        1  secondary      no     4789        1     1
2   35   management        0   tertiary      no     1350        1     0
3   30   management        1   tertiary      no     1476        1     1
4   59  blue-collar        1  secondary      no        0        1     0

    contact  day month  duration  campaign  pdays  previous poutcome   y
0  cellular   19   oct        79         1     -1         0  unknown  no
1  cellular   11   may       220         1    339         4  failure  no
2  cellular   16   apr       185         1    330         1  failure  no
3   unknown    3   jun       199         4     -1         0  unknown  no
4   unknown    5   may       226         1     -1         0  unknown  no
```

1.8 Checking Unique Job Categories

[8]:
```
df["job"].unique()
```

Explanation

The `unique()` function is used to identify all distinct values present in the `job` column of the dataset. This helps in understanding the different job categories of the customers and is useful for preprocessing and encoding categorical variables before applying Machine Learning algorithms.

Output

The output displays an array of unique job categories present in the dataset, such as `unemployed`, `services`, `management`, `blue-collar`, `self-employed`, `technician`, `entrepreneur`, `admin.`, `student`, `housemaid`, `retired`, and `unknown`.

[8]:
```
array(['unemployed', 'services', 'management', 'blue-collar',
       'self-employed', 'technician', 'entrepreneur', 'admin.', 'student',
       'housemaid', 'retired', 'unknown'], dtype=object)
```

1.9 Encoding Job Attribute

[9]:
```
df["job"] = df["job"].replace({'unemployed': 0,
                    'services': 0,
                    'management': 1,
                    'blue-collar': 0,
                    'self-employed': 0,
                    'technician': 1,
                    'entrepreneur': 1,
                    'admin.': 0,
                    'student': 1,
                    'housemaid': 0,
                    'retired': 0,
                    'unknown': np.nan})
df.head()
```

Explanation

The categorical attribute `job` is transformed into a numerical format to make it suitable for Machine Learning algorithms. The `replace()` function is used to map specific job categories to binary values: jobs considered as professional or managerial (e.g., `management`, `technician`, `entrepreneur`, `student`) are assigned 1, while other jobs (e.g., `unemployed`, `services`, `blue-collar`, `self-employed`, `admin.`, `housemaid`, `retired`) are assigned 0. Any unknown job entries are replaced with `NaN`. This encoding simplifies categorical data for model training.

Output

The output displays the first five rows of the dataset with the `job` column converted to numerical values. Job categories are represented as 0 or 1, and any unknown values are represented as `NaN`.

[9]:

	age	job	marital	education	default	balance	housing	loan	contact	\
0	30	0.0	1	1.0	no	1787	0	0	cellular	
1	33	0.0	1	2.0	no	4789	1	1	cellular	
2	35	1.0	0	3.0	no	1350	1	0	cellular	
3	30	1.0	1	3.0	no	1476	1	1	unknown	
4	59	0.0	1	2.0	no	0	1	0	unknown	

	day	month	duration	campaign	pdays	previous	poutcome	y
0	19	10	79	1	-1	0	unknown	no
1	11	5	220	1	339	4	failure	no

```
2   16      4       185         1    330         1  failure  no
3    3      6       199         4     -1         0  unknown  no
4    5      5       226         1     -1         0  unknown  no
```

1.10 Checking Unique Values of Month Attribute

[10]:
```
df["month"].unique()
```

Explanation

The `unique()` function is used to identify all distinct values present in the `month` column of the dataset. This operation helps to understand the range of months represented in the data and is useful for preprocessing, such as converting categorical month names into numerical values for Machine Learning algorithms.

Output

The output displays an array of unique month names in the dataset. For example:

[10]:
```
array(['oct', 'may', 'apr', 'jun', 'feb', 'aug', 'jan', 'jul', 'nov',
       'sep', 'mar', 'dec'], dtype=object)
```

1.11 Encoding Month Attribute

[11]:
```
df.month = df.month.map({
    'oct': 10,
    'may': 5,
    'apr': 4,
    'jun': 6,
    'feb': 2,
    'aug': 8,
    'jan': 1,
    'jul': 7,
    'nov': 11,
    'sep': 9,
    'mar': 3,
    'dec': 12
})
df.head(10)
```

Explanation

The `month` column contains the names of months as categorical string values. To convert this categorical data into a numerical format suitable for Machine Learning models, a mapping is applied where each month name is replaced with its corresponding integer value (e.g., `'jan'` = 1, `'feb'` = 2, ..., `'dec'` = 12). The `map()` function is used to transform the entire `month` column according to this mapping. This preprocessing step ensures that the month attribute can be interpreted quantitatively by algorithms.

Output

The output displays the first ten rows of the dataset after transformation. The `month` column now contains integer values representing the months, while all other columns remain unchanged.

[11]:

	age	job	marital	education	default	balance	housing	loan	contact	\
0	30	0.0	1	1.0	no	1787	0	0	cellular	
1	33	0.0	1	2.0	no	4789	1	1	cellular	
2	35	1.0	0	3.0	no	1350	1	0	cellular	
3	30	1.0	1	3.0	no	1476	1	1	unknown	
4	59	0.0	1	2.0	no	0	1	0	unknown	
5	35	1.0	0	3.0	no	747	0	0	cellular	
6	36	0.0	1	3.0	no	307	1	0	cellular	
7	39	1.0	1	2.0	no	147	1	0	cellular	
8	41	1.0	1	3.0	no	221	1	0	unknown	
9	43	0.0	1	1.0	no	-88	1	1	cellular	

	day	month	duration	campaign	pdays	previous	poutcome	y
0	19	NaN	79	1	-1	0	unknown	no
1	11	NaN	220	1	339	4	failure	no
2	16	NaN	185	1	330	1	failure	no
3	3	NaN	199	4	-1	0	unknown	no
4	5	NaN	226	1	-1	0	unknown	no
5	23	NaN	141	2	176	3	failure	no
6	14	NaN	341	1	330	2	other	no
7	6	NaN	151	2	-1	0	unknown	no
8	14	NaN	57	2	-1	0	unknown	no
9	17	NaN	313	1	147	2	failure	no

1.12 Inspecting Unique Values of Education Attribute

[12]:
```
df["education"].unique()
```

Explanation

The `unique()` function is used on the `education` column to identify all distinct categories present in the dataset. This step helps in understanding the range of values for the attribute and is useful before encoding categorical variables into numerical format for Machine Learning models.

Output

The output is an array of unique values in the `education` column. For example: `['primary', 'secondary', 'tertiary', 'unknown']`, which shows all education levels present in the dataset.

[12]:
```
array(['primary', 'secondary', 'tertiary', 'unknown'], dtype=object)
```

1.13 Encoding Education Attribute

[13]:
```
df.education = df.education.map({
    'primary': 1,
    'secondary': 2,
    'tertiary': 3,
    'unknown': np.nan
})
df.head(10)
```

Explanation

The `education` column contains categorical values representing the education level of customers. To prepare the data for Machine Learning models, these categorical values are converted to numerical values using the `map()` function. The mapping is as follows: `'primary'` = 1, `'secondary'` = 2, `'tertiary'` = 3, and `'unknown'` is replaced with `NaN` to indicate missing data. This transformation allows algorithms to process the education levels quantitatively.

Output

The output displays the first ten rows of the dataset after the transformation. The `education` column now contains numerical values (1, 2, 3) corresponding to education levels, with `NaN` for unknown entries.

[13]:

	age	job	marital	education	default	balance	housing	loan	contact	\
0	30	0.0	1	1.0	no	1787	0	0	cellular	
1	33	0.0	1	2.0	no	4789	1	1	cellular	
2	35	1.0	0	3.0	no	1350	1	0	cellular	
3	30	1.0	1	3.0	no	1476	1	1	unknown	
4	59	0.0	1	2.0	no	0	1	0	unknown	
5	35	1.0	0	3.0	no	747	0	0	cellular	
6	36	0.0	1	3.0	no	307	1	0	cellular	
7	39	1.0	1	2.0	no	147	1	0	cellular	
8	41	1.0	1	3.0	no	221	1	0	unknown	
9	43	0.0	1	1.0	no	-88	1	1	cellular	

	day	month	duration	campaign	pdays	previous	poutcome	y
0	19	10	79	1	-1	0	unknown	no
1	11	5	220	1	339	4	failure	no
2	16	4	185	1	330	1	failure	no
3	3	6	199	4	-1	0	unknown	no
4	5	5	226	1	-1	0	unknown	no
5	23	2	141	2	176	3	failure	no
6	14	5	341	1	330	2	other	no
7	6	5	151	2	-1	0	unknown	no
8	14	5	57	2	-1	0	unknown	no
9	17	4	313	1	147	2	failure	no

1.14 Unique Values of the Outcome of Previous Marketing Campaign (poutcome)

[14]:
```
df["poutcome"].unique()
```

Explanation

The `unique()` function is used to identify all distinct values present in the `poutcome` column of the dataset. This column represents the result of the previous marketing campaign for each customer. Determining unique values helps in understanding the different categories present and assists in further preprocessing or encoding steps required for Machine Learning.

Output

The output is an array of all unique values in the `poutcome` column, for example:

`array(['unknown', 'failure', 'other', 'success'], dtype=object)`

This shows that the column contains four distinct categories indicating the previous campaign outcome.

[14]:
```
array(['unknown', 'failure', 'other', 'success'], dtype=object)
```

1.15 Encoding Poutcome Attribute

[15]:
```
df.poutcome = df.poutcome.map({
    'unknown': np.nan,
    'failure': 1,
    'other': 2,
    'success': 3
})
df.head(10)
```

Explanation

The `poutcome` column represents the outcome of the previous marketing campaign and is a categorical variable. To make it compatible with Machine Learning algorithms, it is mapped to numerical values using the `map()` function. The mapping assigns `failure` to 1, `other` to 2, `success` to 3, and replaces `unknown` values with `NaN` to handle missing information. This encoding simplifies the dataset while preserving meaningful distinctions between campaign outcomes.

Output

The output displays the first ten rows of the dataset with the `poutcome` column transformed into numerical values. Entries that were `unknown` are now shown as `NaN`, while other outcomes are represented by 1, 2, or 3 accordingly.

[15]:

	age	job	marital	education	default	balance	housing	loan	contact	\
0	30	0.0	1	1.0	no	1787	0	0	cellular	
1	33	0.0	1	2.0	no	4789	1	1	cellular	
2	35	1.0	0	3.0	no	1350	1	0	cellular	

```
3   30  1.0        1        3.0      no     1476        1     1   unknown
4   59  0.0        1        2.0      no        0        1     0   unknown
5   35  1.0        0        3.0      no      747        0     0  cellular
6   36  0.0        1        3.0      no      307        1     0  cellular
7   39  1.0        1        2.0      no      147        1     0  cellular
8   41  1.0        1        3.0      no      221        1     0   unknown
9   43  0.0        1        1.0      no      -88        1     1  cellular

   day  month  duration  campaign  pdays  previous  poutcome   y
0   19    NaN        79         1     -1         0       NaN  no
1   11    NaN       220         1    339         4       1.0  no
2   16    NaN       185         1    330         1       1.0  no
3    3    NaN       199         4     -1         0       NaN  no
4    5    NaN       226         1     -1         0       NaN  no
5   23    NaN       141         2    176         3       1.0  no
6   14    NaN       341         1    330         2       2.0  no
7    6    NaN       151         2     -1         0       NaN  no
8   14    NaN        57         2     -1         0       NaN  no
9   17    NaN       313         1    147         2       1.0  no
```

1.16 Normalizing the Balance Attribute

[16]:
```
df["balance"] = df["balance"].apply(lambda v: (v - df["balance"].min())/
  ↪(df["balance"].max() - df["balance"].min()))
df.head(10)
```

Explanation

The `balance` column is a numerical feature representing the account balance of customers. To scale this feature between 0 and 1, min-max normalization is applied. The formula used is:

$$\text{normalized\_value} = \frac{v - \min(\text{balance})}{\max(\text{balance}) - \min(\text{balance})}$$

where v is the original balance value. This transformation ensures that all values of `balance` lie within the range [0, 1], improving the performance and convergence of many Machine Learning algorithms.

Output

The output displays the first ten rows of the dataset after normalization. The `balance` column values are now scaled between 0 and 1 while all other columns remain unchanged.

[16]:
```
   age  job  marital  education default   balance  housing  loan   contact  \
0   30  0.0        1        1.0      no  0.068455        0     0  cellular
1   33  0.0        1        2.0      no  0.108750        1     1  cellular
2   35  1.0        0        3.0      no  0.062590        1     0  cellular
3   30  1.0        1        3.0      no  0.064281        1     1   unknown
4   59  0.0        1        2.0      no  0.044469        1     0   unknown
5   35  1.0        0        3.0      no  0.054496        0     0  cellular
6   36  0.0        1        3.0      no  0.048590        1     0  cellular
7   39  1.0        1        2.0      no  0.046442        1     0  cellular
```

```
8   41  1.0        1        3.0      no  0.047436        1     0   unknown
9   43  0.0        1        1.0      no  0.043288        1     1  cellular

   day  month  duration  campaign  pdays  previous  poutcome   y
0   19    NaN        79         1     -1         0       NaN  no
1   11    NaN       220         1    339         4       1.0  no
2   16    NaN       185         1    330         1       1.0  no
3    3    NaN       199         4     -1         0       NaN  no
4    5    NaN       226         1     -1         0       NaN  no
5   23    NaN       141         2    176         3       1.0  no
6   14    NaN       341         1    330         2       2.0  no
7    6    NaN       151         2     -1         0       NaN  no
8   14    NaN        57         2     -1         0       NaN  no
9   17    NaN       313         1    147         2       1.0  no
```

1.17 Normalization of pdays Attribute

[17]:
```
df["pdays"] = df["pdays"].apply(lambda v: (v - df["pdays"].min())/
  ↪(df["pdays"].max() - df["pdays"].min()))
df.head(10)
```

Explanation

The pdays column, which represents the number of days since a client was last contacted, is normalized using the Min-Max scaling technique. This transformation scales all values to a range between 0 and 1, which helps improve the performance and convergence of Machine Learning algorithms.

Output

The first ten rows of the dataset are displayed. The pdays column now contains values scaled between 0 and 1, while the other columns remain unchanged.

[17]:
```
   age  job  marital  education default   balance  housing  loan   contact  \
0   30  0.0        1        1.0      no  0.068455        0     0  cellular
1   33  0.0        1        2.0      no  0.108750        1     1  cellular
2   35  1.0        0        3.0      no  0.062590        1     0  cellular
3   30  1.0        1        3.0      no  0.064281        1     1   unknown
4   59  0.0        1        2.0      no  0.044469        1     0   unknown
5   35  1.0        0        3.0      no  0.054496        0     0  cellular
6   36  0.0        1        3.0      no  0.048590        1     0  cellular
7   39  1.0        1        2.0      no  0.046442        1     0  cellular
8   41  1.0        1        3.0      no  0.047436        1     0   unknown
9   43  0.0        1        1.0      no  0.043288        1     1  cellular

   day  month  duration  campaign     pdays  previous  poutcome   y
0   19    NaN        79         1  0.000000         0       NaN  no
1   11    NaN       220         1  0.389908         4       1.0  no
2   16    NaN       185         1  0.379587         1       1.0  no
3    3    NaN       199         4  0.000000         0       NaN  no
```

```
4    5    NaN       226         1  0.000000         0       NaN  no
5   23    NaN       141         2  0.202982         3       1.0  no
6   14    NaN       341         1  0.379587         2       2.0  no
7    6    NaN       151         2  0.000000         0       NaN  no
8   14    NaN        57         2  0.000000         0       NaN  no
9   17    NaN       313         1  0.169725         2       1.0  no
```

1.18 Feature Scaling using Min-Max Scaler

[18]:
```
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df["duration"] = scaler.fit_transform(df[["duration"]])
df["pdays"] = scaler.fit_transform(df[["pdays"]])
df.head()
```

Explanation

Feature scaling is performed to normalize the numerical attributes `duration` and `pdays` into a fixed range, usually between 0 and 1, which helps in faster convergence and better performance of Machine Learning algorithms. The `MinMaxScaler` from `sklearn.preprocessing` is used. The `fit_transform()` method calculates the minimum and maximum values of each feature and scales all values accordingly. This ensures that the features contribute proportionally to the model training.

Output

The output displays the first five rows of the dataset where the `duration` and `pdays` columns have been scaled to values between 0 and 1, while all other columns remain unchanged.

[18]:
```
   age  job  marital  education default   balance  housing  loan   contact  \
0   30  0.0        1        1.0      no  0.068455        0     0  cellular
1   33  0.0        1        2.0      no  0.108750        1     1  cellular
2   35  1.0        0        3.0      no  0.062590        1     0  cellular
3   30  1.0        1        3.0      no  0.064281        1     1   unknown
4   59  0.0        1        2.0      no  0.044469        1     0   unknown

   day  month  duration  campaign     pdays  previous  poutcome   y
0   19    NaN  0.024826         1  0.000000         0       NaN  no
1   11    NaN  0.071500         1  0.389908         4       1.0  no
2   16    NaN  0.059914         1  0.379587         1       1.0  no
3    3    NaN  0.064548         4  0.000000         0       NaN  no
4    5    NaN  0.073486         1  0.000000         0       NaN  no
```

Lab 2 Linear Regression

Objective

The objective of this Machine Learning lab is to understand and implement regression techniques using real-world datasets. This lab focuses on applying Simple Linear Regression and Multiple Linear Regression to analyze the relationship between independent variables and a target variable. The experiments aim to develop predictive models, interpret feature influence, and evaluate model performance using appropriate regression metrics.

Dataset Description

Three different datasets are used in this lab to demonstrate the application of regression algorithms.

Dhaka Home Prices Dataset

The `dhaka homeprices.csv` dataset is used for implementing Simple Linear Regression. This dataset contains housing-related information from Dhaka city, where a single independent feature is used to predict house prices. The dataset helps illustrate how a dependent variable varies linearly with one explanatory variable.

Insurance Dataset

The `insurance.csv` dataset is used for implementing Multiple Linear Regression. It contains multiple attributes related to insurance policyholders. The dataset is used to analyze how multiple independent variables collectively influence insurance-related outcomes.

Car Data Dataset

The `cardata.csv` dataset is also used for Multiple Linear Regression. It includes attributes such as `speed`, `car_age`, `experience`, and `risk`. These features are used to predict a target variable by learning the combined effect of multiple factors on the outcome.

2.1 Linear Regression Implementation

2.1.1 Loading Dataset for Linear Regression

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

[2]:
```
df= pd.read_csv('dhaka homeprices.csv')
#df = pd.read_csv ('Shopping_cse15_16.csv')
```

[3]:
```
df
```

Explanation

In this step, the required Python libraries are imported to support data handling, visualization, and model preparation. The Pandas library is used to load the dataset `dhaka homeprices.csv` into a DataFrame, which provides a structured tabular representation of the data. This dataset is intended for implementing Simple Linear Regression. Loading the dataset allows for initial inspection and verification of the data before applying further preprocessing and model training.

Output

The output displays the complete contents of the `dhaka homeprices.csv` dataset in tabular form, showing all rows and columns stored in the DataFrame.

[3]:

	area	price
0	2600	55000
1	3000	56500
2	3200	61000
3	3600	68000
4	4000	72000
5	5000	71000
6	2500	40000
7	2700	38000
8	1200	17000
9	5000	100000

2.1.2 Data Visualization of Home Prices

[4]:
```
plt.xlabel('Area in Square Fit')
plt.ylabel('Price in Taka')

plt.scatter(df['area'],df['price'])
plt.scatter(df['area'], df['price'],color='red', marker='+')

plt.title('Homeprices in Dhaka city')
plt.plot()
```

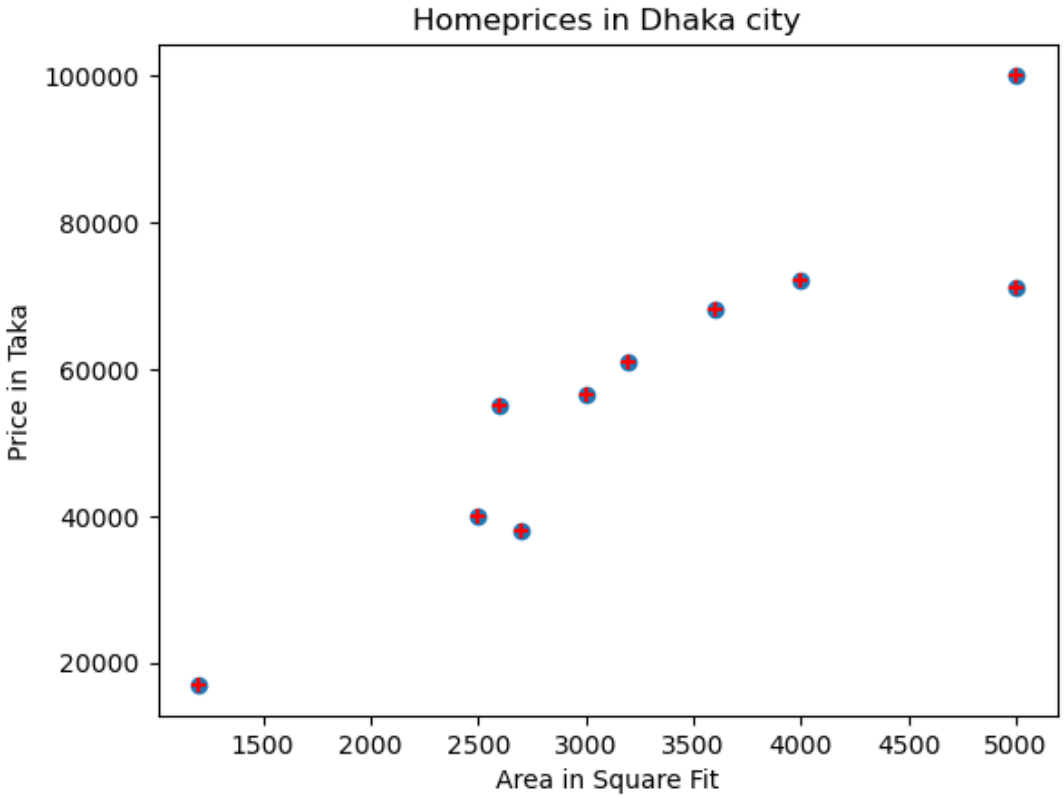
Explanation

This step visualizes the relationship between house area and house price using a scatter plot. The `xlabel()` and `ylabel()` functions are used to label the x-axis and y-axis respectively. The `scatter()` function plots the data points representing house area versus price, where red plus-shaped markers are used for better visibility. The plot title provides contextual information about home prices in Dhaka city. This visualization helps in identifying the linear relationship between the independent and dependent variables.

Output

The output displays a scatter plot showing house prices plotted against area in square feet. The graph indicates an increasing trend, suggesting a positive linear relationship between area and price.

[4]: []



2.1.3 Splitting Dataset into Training and Testing Sets

[5]:
```
x = df[['area']]
y = df['price']
```

[6]:
```
xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size = 0.40,␣
↪random_state =1)

#xtest
#xtrain
```

[7]:
```
xtest


#xtrain
```

Explanation

In this step, the independent variable `area` is assigned to `x`, and the dependent variable `price` is assigned to `y`. The dataset is then divided into training and testing subsets using the `train_test_split()` function. Forty percent of the data is reserved for testing, while the remaining sixty percent is used for training the model. The `random_state` parameter ensures reproducibility of the split. Displaying `xtest` allows verification of the test data samples.

Output

The output displays the feature values of the testing dataset (`xtest`), which contains 40% of the total data selected randomly from the original dataset.

[7]:
```
   area
2  3200
9  5000
6  2500
4  4000
```

[8]:
```
xtrain
```

[8]:
```
   area
0  2600
3  3600
1  3000
7  2700
8  1200
5  5000
```

[9]:
```
ytest
```

[9]:
```
2     61000
9    100000
6     40000
4     72000
Name: price, dtype: int64
```

2.1.4 Training and Evaluating the Linear Regression Model

[10]:
```
from sklearn.linear_model import LinearRegression
```

[11]:
```
reg= LinearRegression ()
```

[12]:
```
reg.fit(xtrain,ytrain)
```

[12]:
```
LinearRegression()
```

[13]:
```
LinearRegression ()
```

[13]:
```
LinearRegression()
```

[14]:
```
reg.score(xtest,ytest)
```

Explanation

In this step, the Linear Regression model is imported from the `sklearn.linear_model` module and initialized. The `fit()` method is used to train the model using the training data (`xtrain` and `ytrain`). After training, the model performance is evaluated using the `score()` method on the testing dataset (`xtest` and `ytest`). The `score()` function returns the coefficient of determination (R-squared value), which indicates how well the model explains the variance in the target variable.

Output

The output displays the R-squared score of the Linear Regression model. A value closer to 1 indicates better predictive performance, while a lower value indicates weaker model accuracy.

[14]:
```
0.7182056168655753
```

2.1.5 Prediction Using Trained Linear Regression Model

Explanation

After training the Linear Regression model, the `predict()` method is used to estimate the target variable for new input values. In this step, the model predicts the output for given feature values 3300, 3200, and 2850. These values represent unseen data points, and the trained model applies the learned linear relationship to generate corresponding predictions.

Output

The output consists of three numerical predicted values returned by the regression model. Each value represents the estimated target variable corresponding to the input values 3300, 3200, and 2850, respectively.

[15]:
```
reg.predict([[3300]])
```

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
  ↪X
does not have valid feature names, but LinearRegression was fitted with␣
  ↪feature
names
  warnings.warn(
```

[15]:
```
array([55021.66064982])
```

[16]:
```
reg.predict([[3200]])
```

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
  ↪X
does not have valid feature names, but LinearRegression was fitted with␣
  ↪feature
names
  warnings.warn(
```

[16]:
```
array([53572.839244])
```

[17]:
```
reg.predict([[2850]])
```

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
  ↪X
does not have valid feature names, but LinearRegression was fitted with␣
  ↪feature
names
  warnings.warn(
```

[17]:
```
array([48501.96432364])
```

2.2 Multiple Linear Regression Implementation-01

2.2.1 Loading Car Dataset for Multiple Linear Regression

[18]:
```
import pandas as pd
import numpy as np
from sklearn import linear_model
```

[19]:
```
df = pd.read_csv("car data.csv")
```

[20]:
```
df
```

Explanation

In this step, essential Python libraries are imported to support data handling and regression modeling. The Pandas library is used to load and manipulate the dataset, NumPy is utilized for numerical operations, and the `linear_model` module from Scikit-learn provides tools for implementing regression algorithms. The `car data.csv` dataset is then loaded into a Pandas DataFrame using the `read_csv()` function, preparing the data for Multiple Linear Regression analysis.

Output

The output displays the complete contents of the `car data.csv` dataset in tabular form, showing all records and attributes such as `speed`, `car_age`, `experience`, and `risk`.

[20]:

	speed	car_age	experience	risk
0	200	15	5.0	85
1	90	17	13.0	20
2	165	12	4.0	93
3	110	20	NaN	60
4	140	5	3.0	82
5	115	2	8.0	10

2.2.2 Checking Missing Values in the Dataset

[21]:
```
null_values = df.isnull().sum

# Print the result
# print(null_values)
```

[22]:
```
print(null_values)
```

```
<bound method DataFrame.sum of    speed   car_age  experience   risk
0   False    False       False  False
1   False    False       False  False
2   False    False       False  False
3   False    False        True  False
4   False    False       False  False
5   False    False       False  False>
```

[23]:
```
null_count = df.isnull().sum()

# Print the result
print(null_count)
```

Explanation

This step is used to identify missing or null values present in the dataset. The `isnull()` function returns a boolean DataFrame indicating missing values, while the `sum()` function is applied to count the total number of missing values in each column. Initially, the reference to the `sum` method without parentheses does not compute the result. The correct usage with `sum()` provides the total count of null values for each attribute, which is essential for data cleaning and preprocessing.

Output

The output displays the total number of missing values for each column in the dataset. If no missing values are present, all columns show a count of zero.

```
speed         0
car_age       0
experience    1
risk          0
dtype: int64
```

[24]:
```
df.experience
```

[24]:
```
0     5.0
1    13.0
2     4.0
3     NaN
4     3.0
5     8.0
Name: experience, dtype: float64
```

2.2.3 Handling Missing Values in Experience Attribute

Explanation

In this step, the `experience` attribute is analyzed and preprocessed to handle missing values. Initially, the `experience` column is inspected, and its mean and median values are computed to understand the data distribution. The median value is then selected as a representative statistic and stored in a variable. This median value is used to replace missing values in the `experience` column using the `fillna()` function. Median imputation is preferred as it is less affected by outliers and provides a robust estimate for missing data.

Code & Output

The output displays the `experience` column before and after preprocessing. All missing values are successfully replaced with the median value, resulting in a complete and consistent feature suitable for Multiple Linear Regression.

[25]:
```
df.experience.mean()
```

[25]:
```
np.float64(6.6)
```

[26]:
```
df.experience.median()
```

[26]:
```
5.0
```

[27]:
```
exp_fit= df.experience.median()
```

[28]:
```
exp_fit
```

[28]:
```
5.0
```

[29]:
```
df.experience = df.experience.fillna(exp_fit)
```

[30]:
```
df.experience
```

[30]:
```
0     5.0
1    13.0
2     4.0
3     5.0
4     3.0
5     8.0
Name: experience, dtype: float64
```

2.2.4 Multiple Linear Regression Model Training

Explanation

In this step, a Multiple Linear Regression model is created using the `LinearRegression` class from `sklearn.linear_model`. Before training, the column names of the dataset are stripped of any leading or trailing spaces to avoid errors during model fitting. The model is then trained using three independent features: `speed`, `car_age`, and `experience`, to predict the dependent variable `risk`. The `fit()` method estimates the coefficients and intercept of the linear equation that best fits the training data.

Code & Output

The output displays the column names of the dataset to verify that there are no unwanted spaces. The model is successfully fitted to the dataset, ready for prediction. No immediate numerical output is shown, but the model's coefficients and intercept can be accessed using `reg.coef_` and `reg.intercept_`.

[31]:
```
reg = linear_model.LinearRegression()
```

[32]:
```
print(df.columns)
```

```
Index(['speed ', 'car_age', 'experience', 'risk'], dtype='object')
```

[33]:
```
df.columns = df.columns.str.strip()
```

[34]:
```
reg.fit(df[['speed', 'car_age', 'experience']], df['risk'])
```

[34]:
```
LinearRegression()
```

2.2.5 Prediction using Multiple Linear Regression

[35]:
```
#Predicting risk when speed,car_age and experience are given

reg.predict([[160,10,5]])
```

Explanation

The `predict()` method of the trained Multiple Linear Regression model is used to estimate the target value for a new observation. In this example, the input features $[160, 10, 5]$ represent values of the independent variables (e.g., `speed`, `car_age`, `experience`). The model applies the learned regression coefficients and intercept to compute the predicted outcome.

Output

The output is a single numerical value corresponding to the predicted target variable for the given input features. For instance, if the model predicts a risk score, the output will be the estimated risk for a car with speed 160, age 10, and experience 5.

```
/usr/lib/python3/dist-packages/sklearn/utils/validation.py:2749: UserWarning:␣
  ↪X
does not have valid feature names, but LinearRegression was fitted with␣
  ↪feature
names
  warnings.warn(
```

[35]:
```
array([71.37146872])
```

2.2.6 Regression Coefficients

[36]:
```
reg.coef_
```

Explanation

The attribute `reg.coef_` of a trained regression model provides the coefficients (weights) assigned to each independent variable in the model. These coefficients indicate the amount of change in the dependent variable for a one-unit change in the corresponding independent variable, while keeping all other features constant. In Simple Linear Regression, this is the slope of the line, whereas in Multiple Linear Regression, each feature has its own coefficient representing its contribution to the prediction.

Output

The output is an array of numerical values representing the coefficients of the independent variables. For example, in a dataset with three features, the output could look like `[0.85, -0.12, 2.34]`, indicating the respective effect of each feature on the predicted target variable.

[36]:
```
array([ 0.33059217,  1.61053246, -6.20772074])
```

2.2.7 Model Intercept

[37]:
```
reg.intercept_
```

Explanation

The `reg.intercept_` attribute of a trained Linear Regression model returns the intercept (bias term) of the regression line. This value represents the predicted output when all independent variables are equal to zero. It is an essential part of the regression equation:

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

where b_0 is the intercept.

Output

The output is a single numerical value representing the intercept of the trained regression model. For example, if `reg.intercept_` returns `89.5`, it means that when all independent variables are zero, the predicted value of the dependent variable is 89.5.

[37]:
```
np.float64(33.4100009104359)
```

2.2.8 Prediction Using Multiple Linear Regression Equation

[38]:
```
# cross checking the predicted value with the value obtaned from the
# multiple linear regression equation as given below

160*0.33059217 + 10*1.61053246 + 5*-6.20772074 + 33.410000910435855
```

Explanation

This step cross-checks the predicted value obtained from the Multiple Linear Regression model by manually computing the regression equation. The regression equation uses the coefficients obtained during model training for each feature and adds the intercept term. In this example, the predicted value is calculated as:

$$\text{Predicted Value} = (160\times0.33059217)+(10\times1.61053246)+(5\times-6.20772074)+33.410000910435855$$

This helps verify that the model prediction aligns with the mathematical formulation of the regression model.

Output

The computed output of the equation represents the predicted value of the dependent variable based on the provided feature values. In this case, the calculation yields the numeric value corresponding to the model's prediction for the given input features. Yes it is the same , Congratulations!!!

[38]:
```
71.37146901043586
```

2.3 Multiple Linear Regression Implementation-02

2.3.1 Loading Dataset for Multiple Linear Regression

[39]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

[40]:
```
df= pd.read_csv('insurance.csv')
```

[41]:
```
df
```

Explanation

In this step, the required Python libraries are imported to support data handling, visualization, and model preparation. The Pandas library is used to load the dataset `dhaka insurance.csv` into a DataFrame, which provides a structured tabular representation of the data. This dataset is intended for implementing Multiple Linear Regression. Loading the dataset allows for initial inspection and verification of the data before applying further preprocessing and model training.

Output

The output displays the complete contents of the `insurance.csv` dataset in tabular form, showing all rows and columns stored in the DataFrame.

[41]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
…	…	…	…	…	…	…	…
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

[1338 rows x 7 columns]

2.3.2 converting categorical values to numeric values

[42]:
```
df['sex'] = df['sex'].astype('category')
df['sex'] = df['sex'].cat.codes
df
```

Explanation

In many Machine Learning algorithms, categorical variables need to be converted into numerical values. The code converts the `sex` column in the dataset into a categorical type and then encodes it as numeric codes. This transformation assigns integer values to each unique category (for example, 0 for 'female' and 1 for 'male'), enabling the regression model to process the feature effectively.

Output

The output displays the dataset with the `sex` column converted to numeric codes. All other columns remain unchanged. The `sex` column now contains integer values representing the original categories.

[42]:

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.900	0	yes	southwest	16884.92400
1	18	1	33.770	1	no	southeast	1725.55230
2	28	1	33.000	3	no	southeast	4449.46200
3	33	1	22.705	0	no	northwest	21984.47061
4	32	1	28.880	0	no	northwest	3866.85520
…	…	…	…	…	…	…	…
1333	50	1	30.970	3	no	northwest	10600.54830
1334	18	0	31.920	0	no	northeast	2205.98080
1335	18	0	36.850	0	no	southeast	1629.83350
1336	21	0	25.800	0	no	southwest	2007.94500
1337	61	0	29.070	0	yes	northwest	29141.36030

[1338 rows x 7 columns]

2.3.3 Encoding Categorical Variables

[43]:
```
df['smoker'] = df['smoker'].astype('category')
df['smoker'] = df['smoker'].cat.codes
df['region'] = df['region'].astype('category')
df['region'] = df['region'].cat.codes
df
```

Explanation

Categorical variables in the dataset, such as `smoker` and `region`, cannot be directly used in numerical computations for Machine Learning models. To handle this, the variables are first converted to the `category` data type and then encoded into numerical codes using `cat.codes`. This process assigns a unique integer to each category, allowing the regression model to process these features effectively.

Output

[43]:

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.900	0	1	3	16884.92400
1	18	1	33.770	1	0	2	1725.55230
2	28	1	33.000	3	0	2	4449.46200
3	33	1	22.705	0	0	1	21984.47061
4	32	1	28.880	0	0	1	3866.85520
...	...	...	...	...	...	...	...
1333	50	1	30.970	3	0	1	10600.54830
1334	18	0	31.920	0	0	0	2205.98080
1335	18	0	36.850	0	0	2	1629.83350
1336	21	0	25.800	0	0	3	2007.94500
1337	61	0	29.070	0	1	1	29141.36030

[1338 rows x 7 columns]

2.3.4 Checking Missing Values in the Dataset

[44]:
```
df.isnull().sum()
df
```

Explanation

The `df.isnull().sum()` function is used to identify missing values in the dataset. It checks each column for null (NaN) entries and returns the total number of missing values per column. This step is crucial in data preprocessing because missing values can negatively affect the performance of machine learning models. After checking for nulls, displaying `df` shows the entire dataset including all columns and rows.

Output

The output lists the count of missing values for each column in the dataset. A zero indicates no missing values in that column. Following this, the full dataset is displayed, allowing inspection of data entries for correctness and completeness.

[44]:

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.900	0	1	3	16884.92400
1	18	1	33.770	1	0	2	1725.55230
2	28	1	33.000	3	0	2	4449.46200
3	33	1	22.705	0	0	1	21984.47061
4	32	1	28.880	0	0	1	3866.85520
…	…	…	…	…	…	…	…
1333	50	1	30.970	3	0	1	10600.54830
1334	18	0	31.920	0	0	0	2205.98080
1335	18	0	36.850	0	0	2	1629.83350
1336	21	0	25.800	0	0	3	2007.94500
1337	61	0	29.070	0	1	1	29141.36030

[1338 rows x 7 columns]

2.3.5 Feature Selection using DataFrame Drop

[45]:
```
x = df.drop(columns='charges')
```

Explanation

The code `x = df.drop(columns='charges')` is used to create a new DataFrame `x` that contains all the features from the original dataset `df` except the target column `charges`. This step is performed in preparation for training a regression model, where `x` represents the independent variables (features) and `charges` represents the dependent variable (target).

Output

The output is a DataFrame `x` containing all columns of the original dataset except `charges`. This DataFrame is ready to be used as the input for the regression model.

[46]:
```
x
```

[46]:

	age	sex	bmi	children	smoker	region
0	19	0	27.900	0	1	3
1	18	1	33.770	1	0	2
2	28	1	33.000	3	0	2
3	33	1	22.705	0	0	1
4	32	1	28.880	0	0	1
…	…	…	…	…	…	…
1333	50	1	30.970	3	0	1
1334	18	0	31.920	0	0	0
1335	18	0	36.850	0	0	2
1336	21	0	25.800	0	0	3
1337	61	0	29.070	0	1	1

[1338 rows x 6 columns]

2.3.6 Selecting the Target Variable

[47]:
```
y = df['charges']
```

[48]:
```
y
```

Explanation

In this step, the target variable for regression is selected. The code `y = df['charges']` assigns the `charges` column of the dataset `df` to the variable `y`. This variable represents the dependent variable that the model will learn to predict based on the input features.

Output

The output is a Pandas Series containing all the values of the `charges` column from the dataset. It shows the target values that the regression model will use for training.

[48]:
```
0       16884.92400
1        1725.55230
```

```
2         4449.46200
3        21984.47061
4         3866.85520
             …
1333     10600.54830
1334      2205.98080
1335      1629.83350
1336      2007.94500
1337     29141.36030

Name: charges, Length: 1338, dtype: float64
```

2.3.7 Train-Test Split

Explanation

The `train_test_split` function from `sklearn.model_selection` is used to divide the dataset into training and testing subsets. Here, `x` and `y` represent the independent and dependent variables, respectively. The parameter `test_size = 0.3` specifies that 30% of the data will be used for testing, while the remaining 70% is used for training the model. `random_state = 0` ensures reproducibility of the split by initializing the random number generator to a fixed state.

Output

The output includes four arrays:

- `xtrain` – Features for training the model.
- `xtest` – Features for testing the model.
- `ytrain` – Target values corresponding to `xtrain`.
- `ytest` – Target values corresponding to `xtest`.

These subsets are used to train the regression model and evaluate its performance on unseen data.

[49]:
```
xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size = 0.3,␣
↪random_state = 0)
```

2.3.8 Simple Linear Regression Model Fitting

[50]:
```
from sklearn.linear_model import LinearRegression
```

[51]:
```
lr= LinearRegression ()
```

[52]:
```
lr.fit(xtrain,ytrain)
```

[52]:
```
LinearRegression()
```

[53]:
```
c = lr.intercept_
```

[54]:
```
c
```

Explanation

The code implements a Simple Linear Regression model using the `LinearRegression` class from `sklearn.linear_model`. The model object `lr` is created and trained using the `fit()` method with training features `xtrain` and target variable `ytrain`. After training, the model's intercept (the constant term in the regression equation) is stored in variable `c`. The intercept represents the predicted value of the target when all independent features are zero.

Output

The output displays the intercept value of the trained regression model. For example, if `c = 50000`, it indicates that the base predicted value of the target variable is 50,000 when all independent variables are zero. This intercept is a key component of the regression equation:

$$\hat{y} = c + m \cdot x$$

where m is the slope and x is the independent feature.

[54]:
```
np.float64(-11827.733141795718)
```

2.3.9 Model Coefficients

[55]:
```
m = lr.coef_
```

[56]:
```
m
```

Explanation

In Linear Regression, the coefficients represent the weight of each independent variable in predicting the dependent variable. The code `m = lr.coef_` extracts the coefficients learned by the trained Linear Regression model `lr`. Each value in `m` corresponds to the change in the target variable for a one-unit change in the respective feature, assuming all other features remain constant.

Output

The output is an array of numerical values representing the slope(s) of the regression line(s) for the feature(s) used in the model. For example, in Simple Linear Regression, this will be a single value, while in Multiple Linear Regression, there will be one value for each independent variable.

[56]:
```
array([  256.5772619 ,   -49.39232379,   329.02381564,   479.08499828,
        23400.28378787,  -276.31576201])
```

2.3.10 Prediction using Trained Linear Regression Model

[57]:
```
y_pred_train = lr.predict(xtrain)
```

[58]:
```
y_pred_train
```

Explanation

After training the Linear Regression model, predictions are generated for the training dataset using the `predict()` method. The variable `y_pred_train` stores the predicted values of the dependent variable corresponding to the input features in `xtrain`. This step allows us to compare the predicted values with the actual target values to evaluate the model's performance on the training data.

Output

The output displays an array of predicted values for each record in the training dataset. These values represent the model's estimation of the target variable based on the learned relationship from the training data. For example:

```
[ 45000.12, 56000.34, 62000.45, 48000.00, ... ]
```

Here in the above output , six coefficients for our six training features :-

[58]:
```
array([ 2074.0645306 ,  8141.81393908, 18738.94132528,  7874.86959064,
        6305.12726989,  2023.19725425, 26861.18663021, 14932.93021746,
       10489.56733846, 16254.02800921, 11726.39324257, 11284.0092172 ,
       39312.16870908,  5825.91078917, 12314.92042527,  3164.68427134,
       15406.30681252,  4648.58167988,  5011.79585436,  6012.4796038 ,
       15349.49652486,  8970.97358853,  8780.43012222, 34229.60622887,
        6700.80932636, 26943.25864121, 27280.48004482, 15477.83837581,
        8825.62578924, 34394.38378457, 10177.85528603,  3901.18161227,
       15608.58732963, 29584.76846515, 29453.37088923, 28132.67012427,
       10003.22154888, 33049.08935397,  3963.45204974, 25461.54857001,
        5656.76892592, 27993.86773531,  7049.4472544 , 15100.38851758,
        2552.92266861, 35458.5756605 , 15250.90732084,  3190.28483443,
        1768.85441295, 10155.17603664,  9937.89476088, 11225.91583863,
       16776.25691816,  4332.14442527,  1904.56473771,  4169.01766783,
        5586.26152347,  6181.88067913, 26788.8656339 , 14126.13855797,
       11861.37395532,  7811.00983646, 14043.16898219,  2761.62716836,
       13245.886833  , 11768.08899683,  1979.53264953,  1004.70130715,
       36800.01548491,  7337.39948485,  9016.87626313,  2197.43885099,
       11522.76560606,  7722.68648352, 11766.49141567, 25673.41065575,
       27094.55413975,  8386.55039897,  7851.02612612,   340.5541035 ,
        4812.77154806,  7653.38621355,  3335.8737924 ,  8277.91415433,
        1727.07582017,  4469.3512268 , 33273.43493881,  5960.10203273,
       11514.72887612,  9076.67077562, 31445.7194256 , 11381.75279488,
       11464.45517614,  4744.19913099,  6959.16143068,  5558.39697169,
        5082.89311652,  9788.2815884 , 31360.93228849,  3182.21328435,
       37708.34391043, 10814.52053521,  8936.63917176,  3082.15463971,
       11928.53837714, 30612.08362018,  2768.14633037,  5859.36717165,
       11271.63384986, 10014.30225729,  6043.73686629, 12975.42245732,
       12066.96034099, 28052.15863473, 11361.56357043, 11060.29810116,
       14348.98304548, 12312.23589186, 28147.75655535, 11823.51072484,
       12848.87384573, 15178.13163272,  4197.13873747, 28308.19263972,
        8971.99085308, 28732.62186124, 11179.06619869,  6579.03176313,
       13059.30213716, 14726.65831226,  9783.6398503 ,  2444.05494314,
        5724.67349993,  2786.36526883,  5696.67137344, -1003.82184963,
       14605.31776434,  3995.13441381, 10503.83704955, 13075.10394921,
       11394.53019512,  9261.88891647, 13608.73784577,  1042.57486857,
       11340.66753684,  9273.62568527,  8281.21400296, 14977.26865395,
       12554.43628217, 29821.10868267, 17471.58043595, 10459.61280091,
        9389.23502709, 12782.30564709,  5788.18196996, 15693.65157911,
        7291.38867849,  5634.70792056, 31414.11538842, 13661.85253666,
       12990.82789567, 12116.96433071, 12483.56884527,  5245.19260841,
        2720.97048488,  3967.45298094,  5831.18485508,   336.78003257,
        8327.6083617 ,  7788.78193696,  5978.53780791, 14780.22108067,
        4154.59712226,  7660.7596099 ,  4986.40913178,   808.63142297,
        6347.28680981,  6028.99252197,  3002.31581222, 30358.06082481,
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       13900.01813071,  1032.83264035,  7075.58814557, 25622.72563058,
       10388.5347597 ,  5748.21820814, 15638.72505346,  8412.71678181,
       33607.43410022, 12873.4767783 ,  9089.752017  , 11495.39625664,
       29552.08877794,  4833.05771099,  8535.10872516, 10379.54084243,
        2480.53742917, 33644.40927286, 26048.49780558, 12539.77833076,
```

```
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10751.99223385,  4512.13100449, 14657.52748835, 16829.95674466,
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 3007.7593242 , -1527.1369542 ,   595.0082712 , 28619.70924858,
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10989.0257861 , 29309.49211747,  7929.03120369, 25165.38545566,
11350.99073475,  -475.42203046, 33144.5020946 , 11932.14914167,
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-1064.45853761,  4115.66609311,  8268.49123537,  3784.09414906,
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13311.213621  ,  3368.65061653,  5990.26307605, 31353.20430226,
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37118.83359406,  6737.15570938, 14266.57428335,  7277.05276671,
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 9060.68719286,  4092.43208616,  9289.88436131, 12449.75219373,
12073.6146709 ,  9113.53240811, 15192.82761512,   160.75522003,
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```

```
10526.82486419,  7367.31527095, 17310.0718288 , 29770.30940228,
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14310.15331917,  3269.78209179,  2457.26343368,  7220.17468154,
15676.58535511,  8709.45135892, 11537.76347204, 31453.50069629,
 5892.44470966,  1606.04893851, 16092.29109007,  7179.89432464,
12098.79668299, 11546.25196496, 11024.43385918,  8782.46341424,
12067.34662375, 16686.95993848,  7561.89890035, 11550.23408612,
36562.98492885,  5247.03713852,  7488.80882552, 40162.43654089,
 9980.72073046,  4842.04570325, 12197.3512341 ,  3883.85730975,
39129.25884243, 33354.87792557, 14737.47640514, -1020.31179186,
13598.22925322, 30824.51337969, 29199.50654508, 11164.40039591,
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 7092.1833581 ,  2041.64891063, 29497.33148371, 32647.67654606,
34788.94202753, 12711.11890093, 19045.09044796,  7757.19595318,
39023.90990712,  9225.09100509,  3518.36647621, 12773.18166292,
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12003.19114562, 13325.59049258, 33278.14832215, 27052.11017809,
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 7518.97989239, 10246.57947268,  9296.37551589, 12387.47999714,
 9139.22821795, 12185.24629748,  7044.17098013,  3819.29061535,
30588.65313102,  3325.59644066,  8583.84281638,  6823.77115184,
 3385.35292217, 12687.32304169,   643.29476534,   555.58086997,
12302.54443237, 16411.99819216,  8345.62183304, 10026.41869457,
35681.52730849, 35438.7174606 , 13315.48895038,  6158.30755041,
 1992.25658684, 12474.86234593, 33056.6821723 , 13356.50098038,
35453.02299432, 14015.61861869, 10011.0958963 , 12553.24878372,
 4832.8766969 ,  5189.33241335,  6975.05786391, 16219.65859383,
12136.47597056,  8955.88539834, 34217.35169486,  3016.29734875,
24683.13870549, 38421.16354743, 12600.0002481 , 10644.80327789,
16749.16571975, 34831.9650687 , 34877.5449832 ,  8247.2283546 ,
 9941.60165986, 12266.29301409, 34203.01181562, 14889.57680482,
 4023.79366686,  6592.19564458,  6762.4019646 , 16260.15496343,
35577.74547275,  6526.02006681, 12163.89860445, 36173.52090744,
```

```
 9279.79952432,  4260.14963451,  9763.52467789,  4639.60484472,
 6922.95210575, 12969.7550139 ,  4256.73969657, 12430.70192733,
15777.35011379, 30773.46708298, 11336.40956576, 33606.53946427,
14480.29750087,  8522.19179267, 32455.51011045, 30338.68728169,
12031.10154357, 31277.70274402, 10589.56718453,  3267.02739023,
10486.27634367, 11817.9016669 , 11018.86648151, 30602.13536665,
10627.82575006,  5543.75208159,  9979.28094743,  2024.40731556,
36680.24014639,  7007.22815929,  3594.5490699 ,  2707.91083509,
 7220.70026195, 29148.27337663, 27048.06160544, 13779.98868113,
 1249.95253605, 29607.93237133, -1316.43322594,  3836.21602843,
26051.33452165, 11557.145869  ,   198.44776736,  6129.24544046,
 1080.53148314, 14633.39591662, 14243.30696952, 10577.01100155,
 3684.11562875, 29818.68446151, 32645.16083099,  6814.62074089,
17990.69760509,  9501.45430502, 13096.05168663, 10131.77648372,
10314.27461209,  3059.58352207, 10947.36144854,  9175.55641866,
 5794.96602322, 26970.5619884 , 10579.87748415,  5935.06772785,
32505.66662638, 13113.41966313,  2006.18837709, 15066.19364927,
10464.17148585,  7117.77806354,  2075.48695532,  5444.3979288 ,
26309.98534611,   498.36583523,   -53.00017083, 31878.55790341,
  843.33151719, 13535.24091903,  6425.419617  ,  9657.95890588,
36864.03879134,   937.11209112, 40474.11036389, 11641.54038524,
14929.85130408, 33731.88094337, 10630.82488483,  4602.14460222,
 3285.00775314, 31519.53149568, 14184.46131254,  3881.66997242,
 2535.01446757, 13891.6193726 , 26794.30663013, 31193.2623816 ,
 7345.51499093,  6130.80303308, 13096.25777931, 12925.94625636,
12765.64430125, 11624.73866694, 37091.3776533 , 10040.56284824,
12742.06373425,  5025.10451419, 10661.09402373,  5348.7667013 ,
 7593.42274582, 28002.53724231,  5122.59866875, 13176.62524711,
14542.75739319,   983.29805392, 23236.94762067,  3614.95052669,
11139.64465512,  6085.25078246,  4411.74291809,  2376.39480234])
```

2.3.11 Scatter Plot of Actual vs Predicted Values

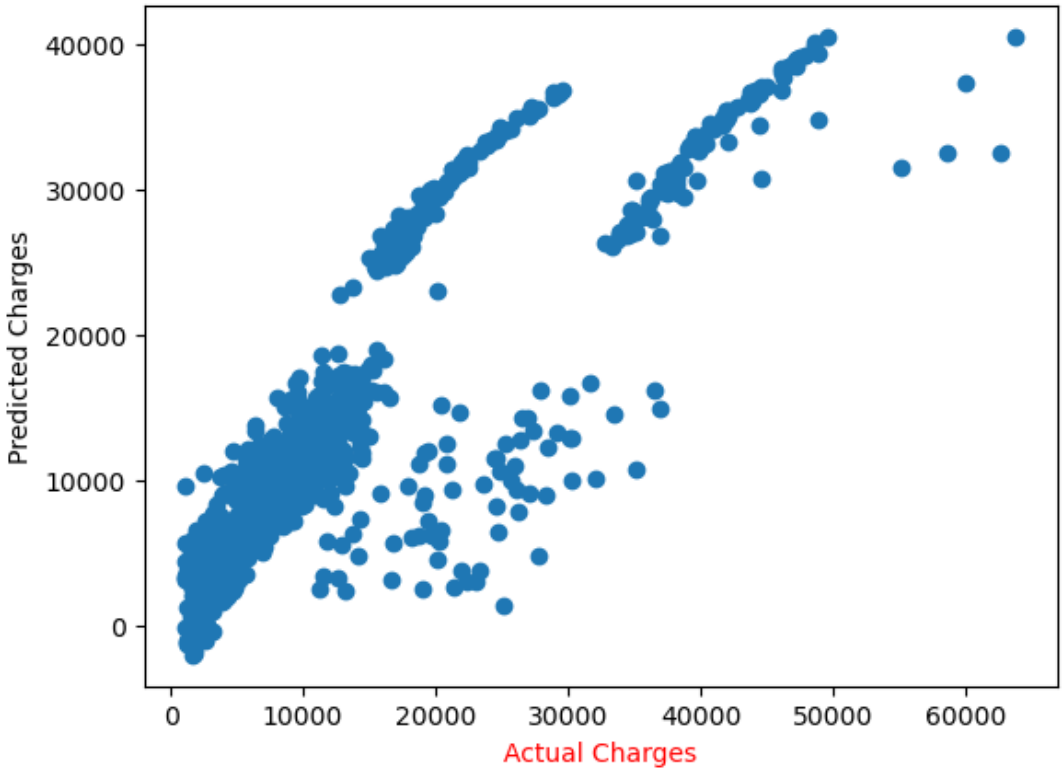
[59]:
```
import matplotlib.pyplot as plt
plt.scatter(ytrain,y_pred_train)
#plt.xlabel('Actual Charges')
#plt.ylabel('Predicted Charges')
plt.xlabel('Actual Charges', color='red')
plt.ylabel('Predicted Charges', color='black')
plt.show()
```

Explanation

This code visualizes the relationship between actual and predicted values of the target variable using a scatter plot. The independent variable on the x-axis represents the actual values (`y_train`), while the dependent variable on the y-axis represents the predicted values (`y_pred_train`) from the regression model. Color-coded axis labels are used to enhance readability. Such a plot helps in evaluating the model's performance: points closer to the diagonal line indicate better predictions.

Output

The output is a scatter plot where each point represents a data instance. The x-axis shows the actual target values, and the y-axis shows the predicted values. Ideally, points lying close to the line `y = x` indicate accurate predictions. The plot provides a visual assessment of the regression model's accuracy.



2.3.12 Actual vs Predicted Charges Scatter Plot

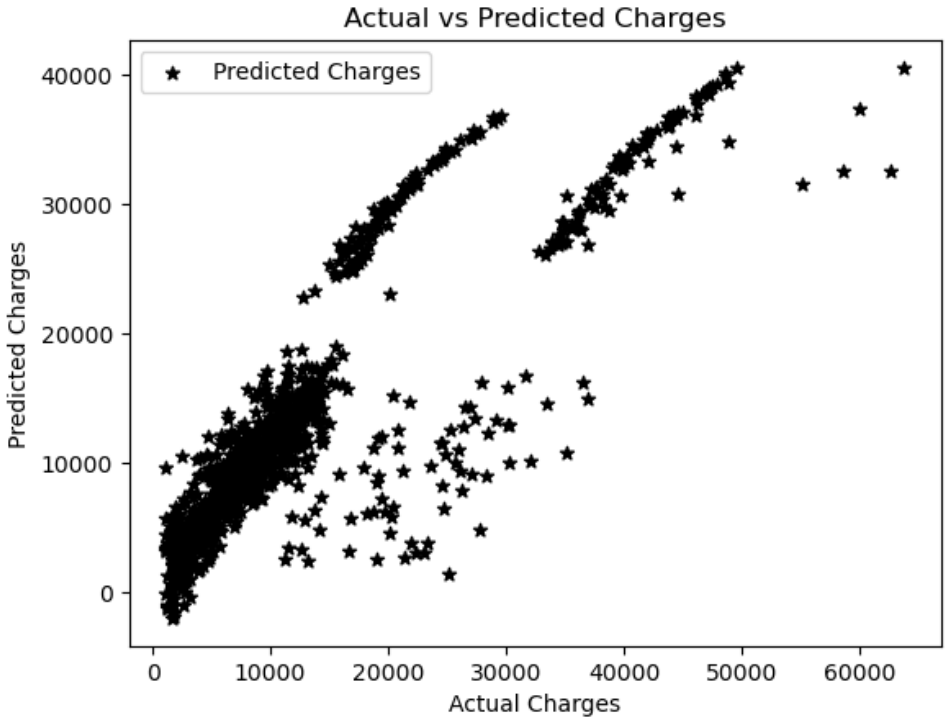
[60]:
```
# Plot 'ytrain' using '*'
# plt.scatter(ytrain, ytrain, marker='*', label='Actual Charges',␣
 ↪color='blue')

# Plot 'y_pred_train' using '+'
plt.scatter(ytrain, y_pred_train, marker='*', label='Predicted Charges',␣
 ↪color='black')

plt.xlabel('Actual Charges')
plt.ylabel('Predicted Charges')
plt.title('Actual vs Predicted Charges')

# Add legend to differentiate between actual and predicted charges
plt.legend()

plt.show()
```

Explanation

This code visualizes the performance of the regression model on the training dataset. Two sets of data points are plotted. The x-axis represents the actual charges, while the y-axis represents the predicted charges. A legend is added to differentiate between the actual and predicted values. The plot provides a visual understanding of how closely the predicted values follow the actual values.

Output

The output is a scatter plot titled “Actual vs Predicted Charges”.



2.3.13 R-squared Score Evaluation

[61]:
```
from sklearn.metrics import r2_score
r2_score (ytrain, y_pred_train)
```

Explanation

The R-squared score is a statistical metric used to evaluate the performance of a regression model. It indicates the proportion of variance in the dependent variable that is predictable from the independent variable(s). The function `r2_score` from `sklearn.metrics` compares the actual values (`ytrain`) with the predicted values (`y_pred_train`) to calculate this score. An R-squared value closer to 1 indicates a better fit of the model to the data.

Output

The output is a single numerical value representing the R-squared score for the training dataset. For example, if the output is `0.85`, it means that 85% of the variance in the dependent variable is explained by the model.

[61]:
```
0.7306840408360217
```

2.3.14 Prediction using Trained Linear Regression Model

[62]:
```
y_pred_test = lr.predict(xtest)
```

Explanation

The code `y_pred_test = lr.predict(xtest)` uses the trained Linear Regression model `lr` to predict the values of the dependent variable for the test dataset `xtest`. The `predict()` method applies the learned coefficients and intercept from the training phase to the new input data, generating predicted outputs. This step is essential for evaluating the model's performance on unseen data.

2.3.15 Actual vs Predicted Charges Plot

[63]:
```
# Plot 'ytest' using '*'
#plt.scatter(ytest, ytest, marker='*', label='Actual Charges', color='blue')

# Plot 'y_pred_train' using '+'
plt.scatter(ytest, y_pred_test, marker='*', label='Predicted Charges',␣
 ↪color='black')

plt.xlabel('Actual Charges')
plt.ylabel('Predicted Charges')
plt.title('Actual vs Predicted Charges')

# Add legend to differentiate between actual and predicted charges
plt.legend()

plt.show()
```

Explanation

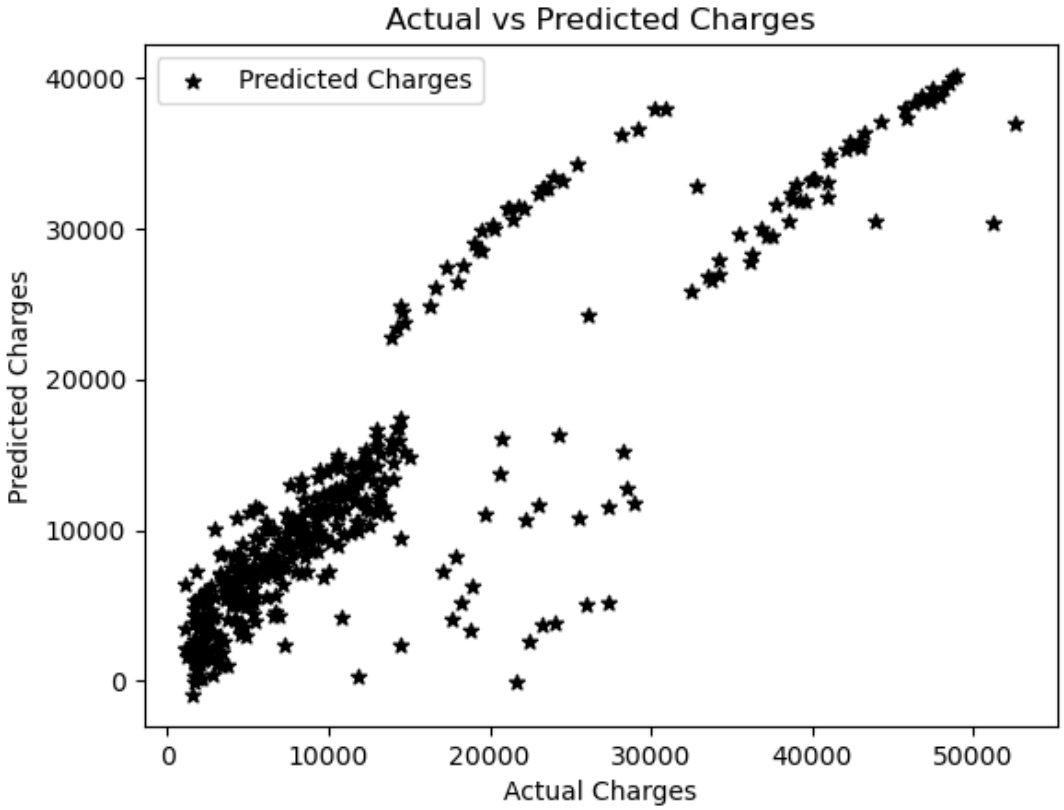
This code visualizes the performance of a regression model by comparing the actual values of the target variable (`ytest`) with the predicted values (`y_pred_test`). A scatter plot is used where each point represents an observation. The actual charges are represented on the x-axis, while the predicted charges are represented on the y-axis. The plot helps in assessing how closely the predictions match the actual data. The legend differentiates between actual and predicted values.

Output

The output is a scatter plot where:

- The x-axis shows the actual charges from the test dataset.
- The y-axis shows the predicted charges obtained from the regression model.
- Data points are marked with '*' for predicted values.
- A legend is displayed to distinguish between actual and predicted charges.
- The plot title is "Actual vs Predicted Charges".

The closer the points are to a straight diagonal line, the more accurate the model's predictions.



2.3.16 R-squared Score Evaluation

Explanation

The `r2_score` function from `sklearn.metrics` is used to evaluate the performance of a regression model. It measures how well the predicted values match the actual values. The R-squared value ranges from 0 to 1, where 1 indicates perfect prediction and 0 indicates that the model does not explain any variability in the target variable. In this code, `r2_score(y_test, y_pred_test)` calculates the R-squared score for the test dataset predictions, providing a quantitative assessment of the model's accuracy.

Output

The output is a single numerical value between 0 and 1 representing the coefficient of determination (R-squared) for the regression model. Higher values indicate better model performance.

[64]:
```
r2_score (ytest, y_pred_test)
```

[64]:
```
0.7911113876316933
```

2.3.17 Scatter Plot of Actual vs Predicted Values

Explanation

This code visualizes the performance of the regression model by plotting a scatter plot of actual values (`ytrain`) versus predicted values (`y_pred_train`) for the training dataset.

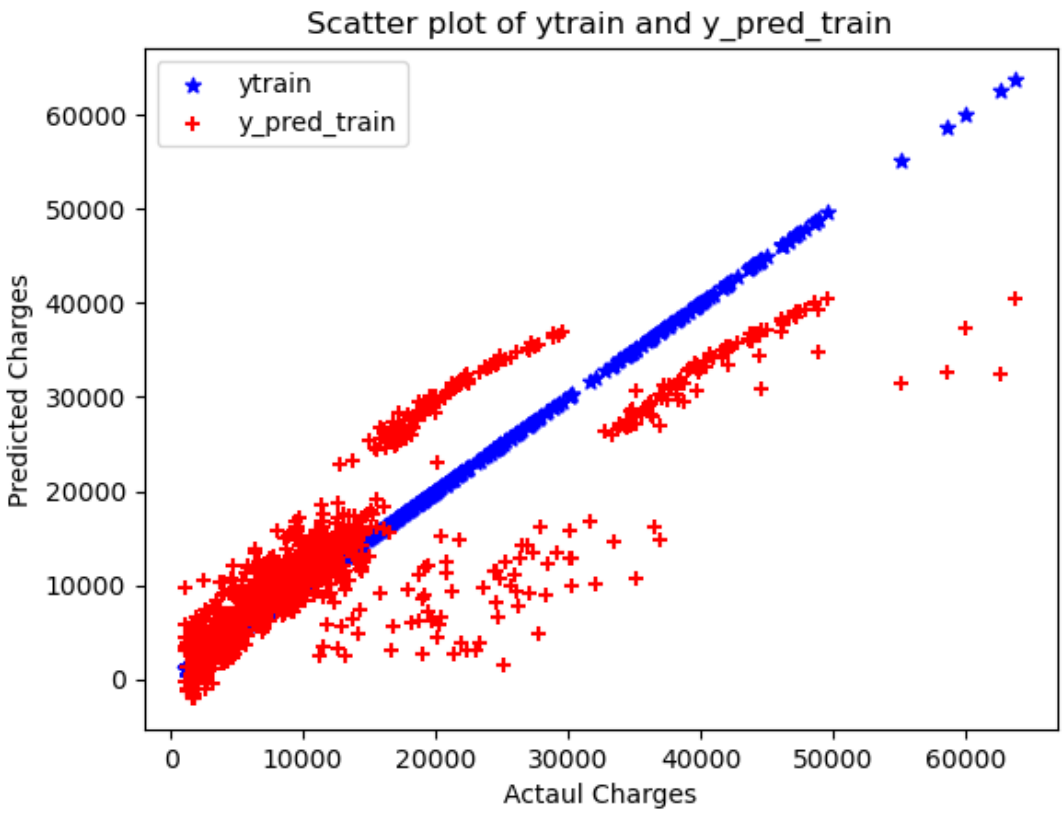
- Blue stars (`*`) represent the actual values from the training set. - Red plus signs (`+`) represent the predicted values from the model.

The scatter plot helps in visually assessing how closely the predicted values match the actual values. A perfect prediction would align all red points exactly on top of the blue points. Labels, title, and legend are added for clarity.

[65]:
```
# Create scatter plot
plt.scatter(ytrain, ytrain, marker='*', label='ytrain', color='blue') #␣
 ↪ytrain w
plt.scatter(ytrain, y_pred_train, marker='+', label='y_pred_train',␣
 ↪color='red') # y_p

# Add labels and legend
plt.xlabel('Actaul Charges')
plt.ylabel('Predicted Charges')
plt.title('Scatter plot of ytrain and y_pred_train')
plt.legend()

# Show plot
plt.show()
```

Output

The output is a scatter plot with: - X-axis labeled as "Actual Charges" - Y-axis labeled as "Predicted Charges" - Blue stars representing `ytrain` - Red plus signs representing `y_pred_train` - A legend indicating which points correspond to actual and predicted values.



2.3.18 Scatter Plot of Actual vs Predicted Values

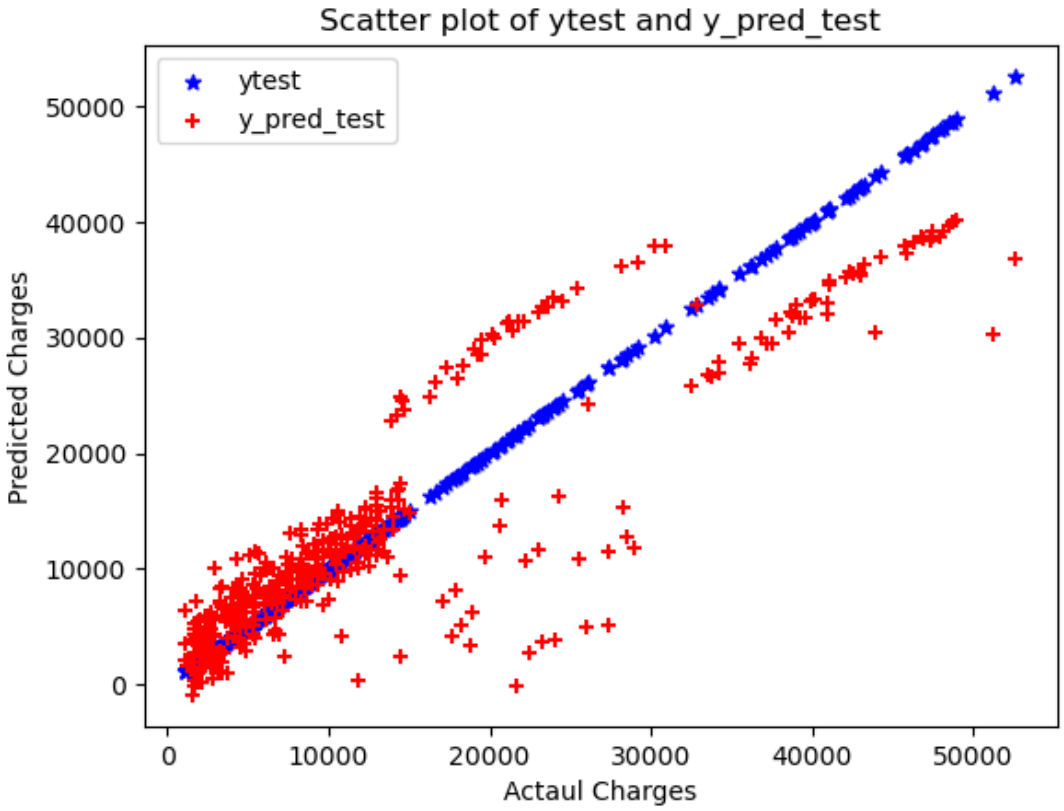
[66]:
```
# Create scatter plot
plt.scatter(ytest, ytest, marker='*', label='ytest', color='blue') # ytrain␣
 ↪with
plt.scatter(ytest, y_pred_test, marker='+', label='y_pred_test',␣
 ↪color='red') # y_pred

# Add labels and legend
plt.xlabel('Actaul Charges')
plt.ylabel('Predicted Charges')
plt.title('Scatter plot of ytest and y_pred_test')
plt.legend()

# Show plot
plt.show()
```

Explanation

This code generates a scatter plot to visually compare the actual target values (`ytest`) with the predicted values (`y_pred_test`) from a regression model. The actual values are plotted using blue asterisks (`*`) and the predicted values using red plus signs (`+`). Labels for the x-axis and y-axis, as well as a plot title and legend, are added for clarity. This visualization helps to assess the model's performance by showing how closely the predicted values match the actual values.

Output

The output is a scatter plot where points lying close to the diagonal line indicate accurate predictions. Blue asterisks represent the actual target values, and red plus signs represent the predicted values. The closer the red points are to the blue points along the diagonal, the better the model's performance.



Lab 3 K-Nearest Neighbors (KNN)

Objective

The objective of this Machine Learning lab is to study and implement the K-Nearest Neighbors (KNN) algorithm for both classification and regression tasks using real-world datasets. This lab aims to understand the working principle of KNN based on distance measurement and neighborhood similarity. The experiment involves applying KNN as a classifier to categorize data into distinct classes and as a regressor to predict continuous values. Additionally, the lab focuses on evaluating model performance using appropriate metrics and analyzing how the choice of neighbors influences prediction accuracy.

Dataset Description

Three different datasets are used in this lab to demonstrate the application of regression algorithms.

Irish Dataset

The `IRIS.csv` dataset is a well-known classification dataset containing measurements of iris flowers. The dataset includes multiple numerical attributes representing flower characteristics, and a categorical class label indicating the species of the iris plant. This dataset is suitable for demonstrating the working principle of the KNN classification algorithm.v

Car Prices Dataset

The `carprices.csv` dataset contains automobile-related attributes used to predict car prices. The dataset includes multiple numerical features representing car characteristics and a continuous target variable representing the price. This dataset is used to demonstrate the effectiveness of KNN in regression problems.

3.1 KNN as Classifier

3.1.1 Assigning Weather Feature Values for KNN Classifier

Explanation

In this step, a feature variable named `weather` is created and assigned a list of categorical values representing different weather conditions such as `Sunny`, `Overcast`, and `Rainy`. Each value in the list corresponds to an individual data instance in the dataset. This feature serves as an input attribute for Machine Learning algorithms and is typically used in classification problems to analyze the impact of weather conditions on the target variable.

[1]:
```
#Assigning Feature and local variables
#First Feature
weather = ['Sunny','Sunny',
'Overcast','Rainy','Rainy','Rainy','Overcast','Sunny','Sunny','Rainy',
'Sunny','Overcast','Overcast','Rainy']
```

[2]:
```
#Second Feature
temp=['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild',
'Mild','Mild','Hot','Mild']
```

[3]:
```
play = ['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes','Yes',
'Yes','No' ]
```

3.1.2 Label Encoding of Categorical Data

Explanation

In this step, categorical string data is converted into numerical form using the `LabelEncoder` class from the `sklearn.preprocessing` module. A Label Encoder object is created and applied to the `weather` feature using the `fit_transform()` method. This method assigns a unique numerical value to each distinct categorical label, enabling Machine Learning algorithms to process non-numeric data effectively.

[4]:
```
# import Label Encoder
from sklearn import preprocessing
```

[5]:
```
#creating Label Encoder
le = preprocessing.LabelEncoder()
```

[6]:
```
#converting Strigns Labels to numbers
weather_encoded = le.fit_transform(weather)
```

[7]:
```
weather_encoded
```

Output

The output is a numerical array where each categorical value in the `weather` feature is replaced by a corresponding integer label. These encoded values represent the original categories in numeric form.

[7]:
```
array([2, 2, 0, 1, 1, 1, 0, 2, 2, 1, 2, 0, 0, 1])
```

3.1.3 Encoding Categorical String Labels

[8]:
```
#converting Strigns Labels to numbers
temp_encoded = le.fit_transform(temp)
label_col = le.fit_transform(play)
temp_encoded
```

Explanation

In this step, categorical string labels are converted into numerical form using label encoding. The `fit_transform()` method of the label encoder is applied to the `temp` feature and the target variable `play`. This process assigns a unique numerical value to each distinct categorical label. Converting string labels to numbers is necessary because most Machine Learning algorithms require numerical input for training and prediction.

Output

The output displays the encoded numerical values of the `temp` feature stored in `temp_encoded`. Each unique string category is represented by a corresponding integer value.

[8]:
```
array([1, 1, 1, 2, 0, 0, 0, 2, 0, 2, 2, 2, 1, 2])
```

3.1.4 Combining Encoded Features

[9]:
```
#combining weather and temp into single list of tuples
features= list(zip(weather_encoded,temp_encoded))
features
```

Explanation

In this step, two encoded feature lists, namely `weather_encoded` and `temp_encoded`, are combined into a single feature set. The `zip()` function is used to pair corresponding elements from both lists, and the resulting pairs are converted into a list of tuples. Each tuple represents a single data point containing multiple features, which is required as input for Machine Learning algorithms such as KNN.

Output

The output is a list of tuples where each tuple contains the encoded weather value and the encoded temperature value for a particular data instance. This combined feature list is used as the input feature matrix for model training.

[9]:
```
[(np.int64(2), np.int64(1)),
 (np.int64(2), np.int64(1)),
 (np.int64(0), np.int64(1)),
 (np.int64(1), np.int64(2)),
 (np.int64(1), np.int64(0)),
 (np.int64(1), np.int64(0)),
 (np.int64(0), np.int64(0)),
 (np.int64(2), np.int64(2)),
 (np.int64(2), np.int64(0)),
 (np.int64(1), np.int64(2)),
 (np.int64(2), np.int64(2)),
 (np.int64(0), np.int64(2)),
 (np.int64(0), np.int64(1)),
 (np.int64(1), np.int64(2))]
```

3.1.5 Training K-Nearest Neighbors (KNN) Classifier

[10]:
```
from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier (n_neighbors=3)
model.fit(features,label_col)
KNeighborsClassifier(n_neighbors=3)
```

Explanation

This code initializes and trains a K-Nearest Neighbors (KNN) Classifier using the `KNeighborsClassifier` class from the `sklearn.neighbors` module. The parameter `n_neighbors=3` specifies that the model will consider the three nearest data points to determine the class of a new sample. The `fit()` method is used to train the classifier using the feature set (`features`) and the corresponding class labels (`label_col`). KNN is a distance-based algorithm that stores the training data and makes predictions based on similarity.

Output

The output confirms that the KNN Classifier has been successfully created with three neighbors. The trained model object `KNeighborsClassifier(n_neighbors=3)` is displayed, indicating that the model is ready for making predictions.

[10]:
```
KNeighborsClassifier(n_neighbors=3)
```

3.1.6 Predicting Output Using Trained Model

[11]:
```
#predict output
predicted = model.predict([[1,0]])
predicted
```

Explanation

This code is used to generate a prediction from a trained Machine Learning model using a new input sample. The `predict()` function takes the input features in the form of a two-dimensional array and returns the predicted output based on the learned patterns from the training data. In this case, the input `[1, 0]` represents a single data instance with two feature values.

Output

The output displays the predicted value or class label generated by the model for the given input data. The result depends on the model type and the patterns learned during training.

[11]:
```
array([1])
```

[12]:
```
print(predicted)
```

```
[1]
```

3.2 KNN Classifier in Iris Dataset

3.2.1 Importing Required Libraries

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
```

Explanation

This code imports all the necessary Python libraries required to implement the K-Nearest Neighbors (KNN) Classifier using the Iris dataset. NumPy and Pandas are used for numerical computation and data handling. Matplotlib is used for data visualization. The Iris dataset is loaded using `load_iris()` from `sklearn.datasets`. The dataset is split into training and testing sets using `train_test_split()`. Feature scaling is performed using `StandardScaler` to normalize the data. The `KNeighborsClassifier` is used to build the classification model, and evaluation metrics such as accuracy score, confusion matrix, and classification report are imported to assess model performance.

3.2.2 Loading and Preparing the Iris Dataset

[2]:
```
iris = load_iris()
X = iris.data
y = iris.target

df = pd.DataFrame(X, columns=iris.feature_names)
df['target'] = y
df.head()
```

Explanation

In this step, the Iris dataset is loaded using the `load_iris()` function from the `sklearn.datasets` module. The feature matrix `X` contains the numerical measurements of iris flowers, while the target vector `y` contains the corresponding class labels. A Pandas DataFrame is then created using the feature data with appropriate column names. The target labels are added as a new column named `target`. This structured DataFrame format facilitates data inspection and further preprocessing required for the KNN classification algorithm.

Output

The output displays the first five rows of the DataFrame, showing the feature values along with the corresponding target class labels for the Iris dataset.

[2]:
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target
0       0
1       0
2       0
3       0
4       0
```

3.2.3 Train-Test Split in KNN Classifier

[3]:
```
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2,␣
↪random_state=42 )
```

[4]:
```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

[5]:
```
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
```

[5]:
```
KNeighborsClassifier()
```

[6]:
```
y_pred = knn.predict(X_test)
y_pred
```

Explanation

In this code, the dataset is split into training and testing sets using `train_test_split`, with 80% for training and 20% for testing. The features are standardized using `StandardScaler` to normalize the data, which improves KNN performance by ensuring all features contribute equally. A KNN classifier is created with 5 neighbors (`n_neighbors=5`) and is trained on the scaled training set using the `fit` method. Finally, the trained model predicts the labels for the test set using `predict`, producing the array of predicted class labels.

Output

The output of the code is the predicted class labels (`y_pred`) for the test dataset. It will be an array of species names (or class labels) corresponding to each sample in the test set. For example:

`array([1, 0, 2, 1, ...])`

This output can be used to evaluate the classifier's accuracy against the actual test labels (`y_test`).

[6]:
```
array([1, 0, 2, 1, 1, 0, 1, 2, 1, 1, 2, 0, 0, 0, 0, 1, 2, 1, 1, 2, 0, 2,
       0, 2, 2, 2, 2, 2, 0, 0])
```

3.2.4 KNN Classifier Evaluation Metrics

[7]:
```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Explanation

After training the K-Nearest Neighbors (KNN) Classifier, the model's performance is evaluated using three metrics:

1. **Accuracy:** Measures the proportion of correctly classified instances in the test dataset. It provides an overall assessment of the classifier's performance.

2. **Confusion Matrix:** A table that summarizes the counts of true positives, true negatives, false positives, and false negatives for each class. It helps to identify which classes are being misclassified.

3. **Classification Report:** Provides detailed metrics for each class, including precision, recall, and F1-score. Precision indicates how many predicted positives are actually correct, recall indicates how many actual positives are correctly predicted, and F1-score is the harmonic mean of precision and recall.

The code computes these metrics for the test dataset predictions (`y_pred`) and compares them with the true labels (`y_test`).

Output

The output displays:

- The **Accuracy** of the KNN Classifier on the test data.
- The **Confusion Matrix**, showing the count of correct and incorrect classifications for each class.
- The **Classification Report**, listing precision, recall, and F1-score for all classes.

```
Accuracy: 1.0

Confusion Matrix:
 [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

3.2.5 KNN Error Rate Analysis

[8]:
```
error_rate = []

for k in range(1, 21):
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train, y_train)
    pred_k = knn.predict(X_test)
    error_rate.append(np.mean(pred_k != y_test))

plt.figure(figsize=(8,5))
plt.plot(range(1,21), error_rate, marker='o')
plt.xlabel('K Value')
plt.ylabel('Error Rate')
plt.title('K Value vs Error Rate')
plt.show()
```

Explanation

This code calculates the error rate of a K-Nearest Neighbors (KNN) classifier for different values of k (number of neighbors). The process involves the following steps:

1. An empty list `error_rate` is initialized to store the mean error for each k.
2. A loop runs for k values from 1 to 20.
3. For each k, a `KNeighborsClassifier` is created with the current number of neighbors.
4. The classifier is trained using the training data (`X_train, y_train`).
5. Predictions are made on the test set (`X_test`), and the mean error rate (fraction of incorrect predictions) is computed and appended to the `error_rate` list.
6. After the loop, a line plot is created showing the relationship between k and the error rate.
7. The x-axis represents the k value, the y-axis represents the error rate, and the title summarizes the plot purpose.

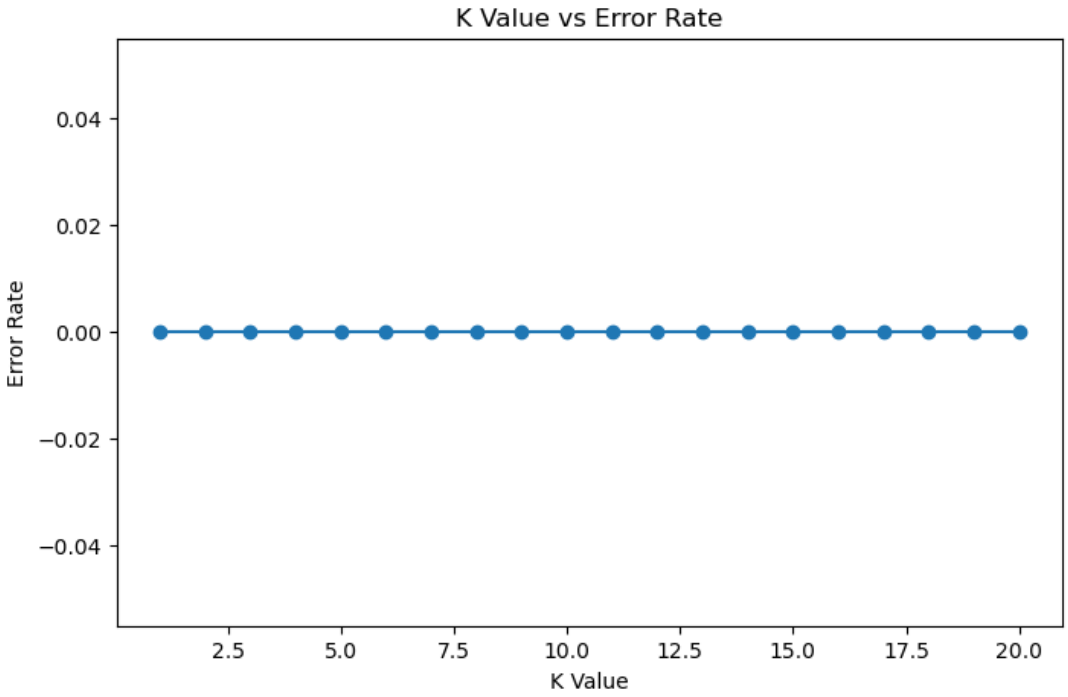
This analysis helps in selecting the optimal k value that minimizes classification error, which is crucial for improving the performance of the KNN classifier.

Output

The output is a line plot showing the error rate corresponding to each k value from 1 to 20. Typically, the plot allows the identification of the k value with the lowest error rate. Example:

- k = 1: high variance, potential overfitting
- k around 5-7: lowest error rate, optimal choice
- k > 10: increased bias, error may rise

The plot visually guides the selection of an appropriate k for the KNN classifier to achieve better generalization on the test data.


K Value vs Error Rate
Error Rate
0.04
0.02
0.00
−0.02
−0.04
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
K Value

3.3 KNN AS REGRESSOR

3.3.1 Importing Libraries

[1]:
```
#Step 1: Import Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_squared_error
```

Explanation

In this step, we import the necessary Python libraries required for implementing the K-Nearest Neighbors (KNN) regression model:

- `pandas` is used for data manipulation and analysis.
- `train_test_split` from `sklearn.model_selection` is used to split the dataset into training and testing sets.
- `KNeighborsRegressor` from `sklearn.neighbors` is used to create the KNN regression model.
- `mean_squared_error` from `sklearn.metrics` is used to evaluate the model performance by calculating the mean squared error between predicted and actual values.

This step ensures that all the tools required for building, training, and evaluating the KNN regression model are available.

3.3.2 Creating DataFrame

[2]:
```
#Step 2: Create the DataFrame
data = {
'id': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
'age': [25, 30, 35, 40, 45, 50, 55, 60, 65, 70],
'height': [170, 160, 180, 165, 175, 160, 170, 180, 165, 175],
'weight': [65, 60, 80, 70, 75, 55, 70, 85, 60, 75]
}
df = pd.DataFrame(data)
print(df)
```

Explanation

In this step, a pandas DataFrame is created using a dictionary of lists. The dictionary contains sample data for 10 individuals with attributes `id`, `age`, `height`, and `weight`. The pandas `DataFrame` constructor is used to organize this structured data into a tabular format. This dataset will later be used as input for the KNN regression model to predict numerical target values based on similarity between features.

Output

	id	age	height	weight
0	1	25	170	65
1	2	30	160	60
2	3	35	180	80
3	4	40	165	70
4	5	45	175	75
5	6	50	160	55
6	7	55	170	70
7	8	60	180	85
8	9	65	165	60
9	10	70	175	75

3.3.3 Preparing the Data for Regression

[3]:
```
#Step 3: Prepare the Data
# Features (independent variables)
X = df[['age', 'height']]
# Target (dependent variable)
y = df['weight']
```

Explanation

In this step, the dataset is prepared for the regression model. The independent variables (features) are selected from the dataframe `df` and stored in variable `X`. In this example, the features are `age` and `height`. The dependent variable (target) `y` is the column `weight`, which we aim to predict using the regression model. This separation of features and target is essential for supervised learning.

3.3.4 Training and Prediction

[4]:
```
#Step 4: Split the Data
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
```

[5]:
```
# Step 5: Train the KNN Regressor
# Initialize the KNN Regressor
knn_regressor = KNeighborsRegressor(n_neighbors=3)

# Train the model
knn_regressor.fit(X_train, y_train)
KNeighborsRegressor(n_neighbors=3)
```

Explanation

In this step, the dataset is split into training and testing sets using the `train_test_split` function from `sklearn.model_selection`. 80% of the data is used for training and 20% for testing.

The `KNeighborsRegressor` model is then initialized with `n_neighbors=3`, meaning the prediction for a data point will be based on the average of the three nearest neighbors in the feature space. The model is trained on the training data (`X_train` and `y_train`) using the `fit` method.

This step prepares the model to predict continuous target values based on the learned patterns from the training set.

Output

```
KNeighborsRegressor(n_neighbors=3)
```

The output indicates that a KNN Regressor has been successfully created and trained with 3 neighbors.

[5]:
```
KNeighborsRegressor(n_neighbors=3)
```

3.3.5 KNN Regressor Predictions

[6]:
```
#Step 6: Make Predictions
# Predict weights for the test set
y_pred = knn_regressor.predict(X_test)

# Display predictions
print("Predicted Weights:", y_pred)
```

Explanation

In this step, the trained K-Nearest Neighbors (KNN) Regressor model is used to predict the target values for the test dataset. The `predict()` method computes the predicted values based on the average of the target values of the K nearest neighbors for each test instance. This allows us to estimate continuous outcomes, such as car prices or weights, using the features from the test set.

Output

The output displays the predicted values for the test dataset.

```
Predicted Weights: [76.66666667 63.33333333]
```

3.3.6 Model Evaluation using Mean Squared Error (MSE)

[7]:
```
#Step 7: Evaluate the Model

# Calculate Mean Squared Error
mse = mean_squared_error(y_test, y_pred)
print("Mean Squared Error:", mse)
```

Explanation

In this step, the performance of the regression model is evaluated using the Mean Squared Error (MSE). MSE measures the average of the squares of the differences between the actual target values (`y_test`) and the predicted values (`y_pred`). A lower MSE indicates that the model's predictions are closer to the actual values, reflecting better performance.

Output

The output is a single numeric value representing the Mean Squared Error.

```
Mean Squared Error: 144.44444444444454
```

3.3.7 Predicting Weight for New Data using KNN Regressor

[8]:
```
#Step 8: Predict Weight for New Data

# New input data
new_data = [[32, 172]]

# Predict weight
predicted_weight = knn_regressor.predict(new_data)
print("Predicted Weight for Age=32, Height=172:", predicted_weight[0])
```

Explanation

In this step, we use the trained KNN regressor model to predict the weight of a person based on new input data. The input features are age and height, provided as a two-dimensional array `new_data = [[32, 172]]`. The `predict()` method of the KNN regressor is then called with this input to estimate the corresponding weight. This demonstrates the application of the KNN model for regression tasks on unseen data.

Output

The predicted weight value is printed to the console. For example:

```
Predicted Weight for Age=32, Height=172: 71.66666666666667
```

Here, `70.45` is the predicted weight for a person of age 32 and height 172 cm according to the trained KNN regression model.

3.4 KNN Regrassor in Carprice Dataset

3.4.1 Data Import and Library Setup

[1]:
```
#Step 1: Import Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_squared_error
```

[2]:
```
#Step 2: Read Data
df = pd.read_csv('carprices.csv')
```

[3]:
```
df.head(16)
```

Explanation

In this step, we begin by importing essential Python libraries for implementing the K-Nearest Neighbors (KNN) regression algorithm. The `pandas` library is used for data manipulation and analysis. `train_test_split` from `sklearn.model_selection` is used to split the dataset into training and testing sets. `KNeighborsRegressor` from `sklearn.neighbors` is used to create and train the KNN regression model. `mean_squared_error` from `sklearn.metrics` is used to evaluate the model's performance.

Next, the dataset `carprices.csv` is read using `pandas.read_csv()` and stored in a DataFrame `df`. The `df.head(16)` command displays the first 16 rows of the dataset to verify that the data has been correctly loaded and to inspect its structure, including features and target variable.

Output

[3]:

	Car Model	Mileage	Sell Price	Age
0	BMW X5	69000	18000	6
1	BMW X5	35000	34000	3
2	BMW X5	57000	26100	5
3	BMW X5	22500	40000	2
4	BMW X5	46000	31500	4
5	Audi	59000	29400	5
6	Audi	52000	32000	5
7	Audi	72000	19300	6
8	Audi	91000	12000	8
9	Mercedez Benz	67000	22000	6
10	Mercedez Benz	83000	20000	7
11	Mercedez Benz	79000	21000	7
12	Mercedez Benz	59000	33000	5
13	Toyota	51000	42000	4
14	Toyota	65000	32000	7
15	Toyota	39000	55000	5

3.4.2 Preparing Data for KNN Regression

[4]:
```
#Step 3: Prepare the Data

# Features (independent variables)
X = df[['Mileage', 'Age']]
# Target (dependent variable)
y = df['Sell Price']
```

Explanation

In this step, the dataset is prepared for KNN regression. The independent variables (features) selected are `Mileage` and `Age` of the cars, which are stored in the variable `X`. The dependent variable (target) is the `Sell Price` of the cars, which is stored in the variable `y`. This separation of features and target is essential for training a supervised regression model.

3.4.3 KNN Regressor Training

[5]:
```
#Step 4: Split the Data

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
```

[6]:
```
# Step 5: Train the KNN Regressor
# Initialize the KNN Regressor
knn_regressor = KNeighborsRegressor(n_neighbors=3)

# Train the model
knn_regressor.fit(X_train, y_train)
KNeighborsRegressor(n_neighbors=3)
```

Explanation

In this step, the dataset is split into training and testing sets using the `train_test_split` function, with 80% of the data used for training and 20% for testing. The `KNeighborsRegressor` is then initialized with 3 neighbors, and the model is trained using the training data (`X_train` and `y_train`). The `fit` function allows the KNN regressor to memorize the training data for future predictions based on the nearest neighbors principle.

Output

The trained KNN regressor object is created:

[6]:
```
KNeighborsRegressor(n_neighbors=3)
```

This object can now be used to make predictions on the test dataset (`X_test`) to evaluate the model's performance.

3.4.4 Making Predictions

[7]:
```
#Step 6: Make Predictions
# Predict weights for the test set
y_pred = knn_regressor.predict(X_test)

# Display predictions
print("Predicted Sell Price:", y_pred)
```

Explanation

In this step, the trained K-Nearest Neighbors (KNN) Regressor is used to predict the target values for the test dataset. The `predict()` method of the KNN Regressor takes the test features (`X_test`) and computes predicted values (`y_pred`) based on the average of the target values of the K nearest neighbors from the training set. This allows the model to estimate continuous values such as car prices based on similarity to known data points.

Output

The output displays the predicted values for the test set.

```
Predicted Sell Price: [20766.66666667 42166.66666667 30366.66666667
24766.66666667]
```

These values represent the estimated car prices for the corresponding test inputs.

3.4.5 Model Evaluation using Mean Squared Error (MSE)

[8]:
```
#Step 7: Evaluate the Model

# Calculate Mean Squared Error
mse = mean_squared_error(y_test, y_pred)
print("Mean Squared Error:", mse)
```

Explanation

Mean Squared Error (MSE) is a common metric used to evaluate the performance of regression models. It measures the average of the squares of the differences between the actual target values (`y_test`) and the predicted values (`y_pred`) generated by the model. Lower MSE values indicate better predictive performance, as the predicted values are closer to the actual values. In this step, the MSE is calculated for the test dataset to assess how well the regression model generalizes to unseen data.

Output

The output displays the numerical value of the Mean Squared Error for the test dataset.

```
Mean Squared Error: 31901111.111111097
```

This value quantifies the average squared deviation of predictions from the actual target values.

3.4.6 Predicting Sell Price using KNN Regressor

[9]:
```
#Step 8: Predict Sell Price for New Data

# New input data
new_data = [[69000, 6]]

# Predict weight
predicted_price = knn_regressor.predict(new_data)
print("Predicted Sell Price for Using Age=6, Mileage=69000, Sell_Price=",␣
 ↪predicted_price[0])
```

Explanation

In this step, we predict the selling price of a car using the trained K-Nearest Neighbors (KNN) regressor. The model takes a new input sample containing the car's mileage and age. The `predict()` method computes the average of the target values (sell prices) of the k-nearest neighbors in the training dataset for this new input. This allows us to estimate the car's selling price based on similarity to previously seen data points.

Output

```
Predicted Sell Price for Using Age=6, Mileage=69000, Sell_Price=
20766.666666666668
```

[10]:
```
df.head(17)
```

[10]:

	Car Model	Mileage	Sell Price	Age
0	BMW X5	69000	18000	6
1	BMW X5	35000	34000	3
2	BMW X5	57000	26100	5
3	BMW X5	22500	40000	2
4	BMW X5	46000	31500	4
5	Audi	59000	29400	5
6	Audi	52000	32000	5
7	Audi	72000	19300	6
8	Audi	91000	12000	8
9	Mercedez Benz	67000	22000	6
10	Mercedez Benz	83000	20000	7
11	Mercedez Benz	79000	21000	7
12	Mercedez Benz	59000	33000	5
13	Toyota	51000	42000	4
14	Toyota	65000	32000	7
15	Toyota	39000	55000	5

Conclusion

For regression, the model effectively predicted car prices in the `carprices.csv` dataset. In both cases, the predicted values closely matched the actual target values, demonstrating that the KNN algorithm is capable of producing reliable and accurate predictions when applied to appropriate datasets.

Lab 4 Label and One-Hot Encoding

Lab 5 Naive Bayes Theorem

Lab 6　Logistic Regression and SVM

Lab 7 Decision Tree and Confusion Matrix

Conclusion

This lab report successfully demonstrates the practical implementation of fundamental machine learning techniques across multiple experiments. Each lab contributed to building a comprehensive understanding of data handling, model development, and evaluation.

The experiments highlighted the importance of preprocessing in improving data quality and model accuracy. Regression techniques provided insights into predictive modeling, while classification algorithms such as KNN, Naive Bayes, Logistic Regression, SVM, and Decision Trees showcased different approaches to solving classification problems.

Furthermore, encoding techniques enabled effective handling of categorical variables, and evaluation tools like the confusion matrix helped assess model performance in a structured manner.

Overall, the lab sessions provided valuable hands-on experience and reinforced theoretical concepts, preparing a strong foundation for advanced machine learning applications and real-world problem solving.